

PROJECTION-TRANSPORT STEERING: CONDITIONAL CONTROL OF REASONING BEHAVIOR IN LLMs

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ABSTRACT

Activation steering edits a language model’s residual stream at inference time, but most edits act on every input, so an edit strong enough to remove a behavior also changes inputs that never showed it. We propose Projection-Transport Steering (PTS), a framework that splits steering into an action and a decision. The action is the optimal-transport map of a low-dimensional behavioral projection onto a target law, the unique minimum-displacement edit that leaves the rest of the residual stream untouched. The decision is a likelihood-ratio test on a learned detection statistic. The selectivity of the composed operator decomposes exactly into the decision’s error rates and the action’s conversion rates, so one ungated run predicts gated selectivity and unsteered answers give the selectivity ceiling of a decision. On overconfident multiple-choice reasoning with Gemma-2-2B, PTS is the most selective of eight methods on held-out questions and lowers calibration error at non-inferior accuracy. On free-text toxicity in three model families and two held-out prompt sets, PTS is more selective than refusing wherever its own decision fires. On Gemma, an edit along the displacement of the quantile transport map, selected on the tuning split, is more selective than a shift of equal mean displacement.

1 INTRODUCTION

Many high-level behaviors of large language models appear to be carried by directions in their activations. Activation steering methods (Turner et al., 2023; Panickssery et al., 2024; Zou et al., 2023; Li et al., 2023) read such a direction off contrastive examples and add it to the residual stream at inference time, at the cost of one vector addition per token and no retraining. However, these methods apply the same edit to every input, so the strength needed to remove a behavior where it occurs also changes the inputs where it does not: a refusal vector at a useful strength raises refusal on harmful and harmless prompts alike (Arditi et al., 2024).

The difficulty is sharpest for a behavior that a model shows on only some inputs, such as overconfidence in multiple-choice reasoning, where the answers to change and the answers to protect are both confident, one wrong and one right, and the two states overlap on the model’s confidence direction. On Gemma-2-2B, our main model, a global shift along it strong enough to remove every overconfident-wrong answer also removes every confident-right one (Figure 1a).

In this paper we introduce Projection-Transport Steering (PTS), which splits a steering operator into an *action* that sets how to edit and a *decision* that sets which inputs to edit. The action is a transfer function on the projection of the residual stream onto a behavioral subspace, a few directions read off the model’s own labeled outputs. The decision is a likelihood-ratio test on a detection statistic fitted to tell the target state from the protected one, and the action runs only where the test fires. Our method is based on the observation that this statistic, read from the full activation, separates the two confident states better than the coordinate the action moves (Sec. 5). Table 1 places published operators in this frame.

Given the subspace and a target law for the steered projection, the action is determined: the optimal-transport map between the two laws is the unique minimum-displacement edit that leaves the orthogonal complement untouched (Theorem 1). Additive steering, ablation and clamping are this map for

method	transfer function $g(p)$	decision	cost / token
ActAdd, CAA (Turner et al., 2023; Panickssery et al., 2024)	$p + \alpha$	none	$\Theta(d)$
ITI (Li et al., 2023)	$p + \alpha$ on probe-selected heads	none	$\Theta(d)$
RepE (Zou et al., 2023)	$p + \alpha(x)$	none	$\Theta(d)$
directional ablation (Arditi et al., 2024)	0	none	$\Theta(d)$
SAE clamping (Templeton et al., 2024)	c	none	$\Theta(d d_{\text{sae}})$
MiMiC (Singh et al., 2024)	full-space affine map	none	$\Theta(d^2)$
Linear-AcT (Rodriguez et al., 2025a)	$(1-\lambda)p + \lambda(\omega p + \beta)$, per neuron	none	$\Theta(d)$
CAST (Lee et al., 2025)	$p + \alpha$	prompt similarity	$\Theta(d)$
PCHI (Li et al., 2026)	learned head rescaling	probe at the readout	$\Theta(d)$
PTS (ours)	$S(p)$, the transport map to ν	likelihood-ratio test	$\Theta(kd)$

Table 1: Published steering operators differ in their transfer function and in whether a decision gates it. Each row applies a transfer function g to a k -dimensional projection p of the residual stream; PTS derives g from a target law ν (Theorem 1) and accounts for its decision exactly (Proposition 1).

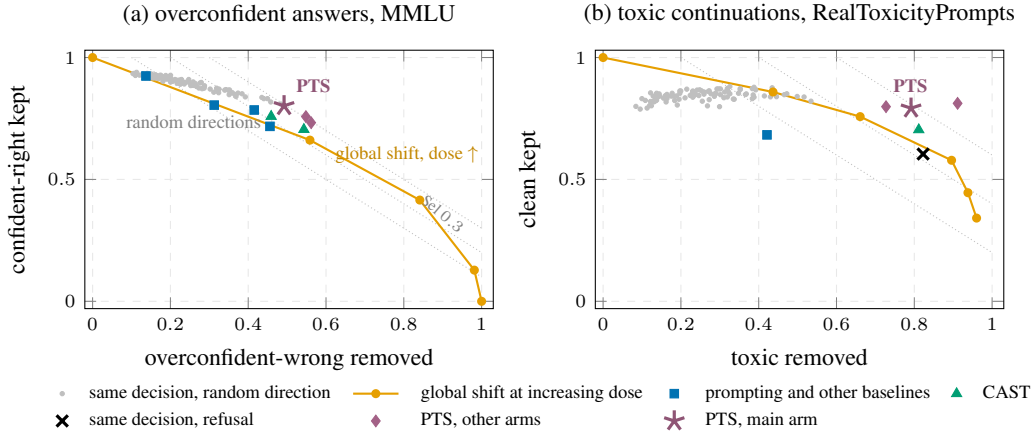


Figure 1: PTS removes more of the target behavior than the global shift at the same retention. Held-out results on Gemma for overconfident answers (a) and toxic continuations (b); up and to the right is better, and dotted lines mark equal selectivity. The orange curve is the global shift at increasing dose, gray points are random directions sent through the PTS decision, and the star is the main PTS arm (post-hoc shift in a, scaled transport displacement in b); other markers are tuned methods.

particular targets, and on a one-dimensional projection the map to any target is a monotone quantile map. The decision enters through an exact identity: the selectivity of a conditional edit, the share of target outputs it removes minus the share of protected outputs it loses, equals the decision’s true-positive rate times the rate at which the action converts flagged targets, minus its false-positive rate times the rate at which it converts flagged protected outputs (Proposition 1). The identity predicts gated selectivity from one ungated run and gives the ceiling of a decision from unsteered answers (Theorem 2); with Eq. (3) it also states when steering beats overwriting every flagged output.

Our main contributions are:

- The framework and its theory (Sec. 3), with a check of the decomposition: over 240 tuning-split configurations per model, it predicts gated selectivity with correlations of 0.97 and 0.91 (Sec. 5.2).
- Held-out evidence on overconfident reasoning, with every configuration committed before the held-out run: on 3,000 MMLU questions for the main model, PTS is the most selective of eight methods, and its direction is more selective than each of a hundred random directions with the same decision and dose (Sec. 5).
- Free-text control of toxic continuation in three model families, where PTS beats overwriting every flagged continuation with a refusal, in every family and on two disjoint prompt sets; on the main model, the edit along the transport displacement selected on the tuning split is more selective than a shift of equal mean displacement (Sec. 6, Figure 1b).
- In the setting of PCHI, a PTS shift along one direction is more selective than PCHI’s learned rescaling of attention heads under the same probe (Sec. 6).

2 RELATED WORK

Activation steering. ActAdd (Turner et al., 2023) adds the activation difference of one contrast prompt pair to the residual stream, and CAA (Panickssery et al., 2024) averages it over many pairs. ITI (Li et al., 2023) shifts attention heads selected by linear probes, RepE (Zou et al., 2023) reads and writes representation directions, directional ablation (Arditi et al., 2024) removes one direction from every activation, and SAE clamping (Templeton et al., 2024) fixes one sparse feature. AxBench (Wu et al., 2025) finds prompting stronger than such edits, and reliability studies (Tan et al., 2024; Braun et al., 2025) report that their effect varies widely across inputs. These operators act on every input with a transfer function chosen in advance; PTS derives the transfer function from a target law (Theorem 1).

Transport maps. MiMiC (Singh et al., 2024) applies to the full activation the affine map that matches the mean and covariance of two activation distributions, the Gaussian optimal-transport map. Linear-AcT (Rodriguez et al., 2025a) fits a univariate affine transport map to each neuron, interpolated by a strength $\lambda \in [0, 1]$, and writes earlier edits as its special cases; LinEAS (Rodriguez et al., 2025b) learns such maps across layers end to end. COAST (Nguyen et al., 2026) moves each activation, at fixed norm, to a fixed cosine with a target direction at the smallest second-moment-weighted displacement, which in our terms transports that cosine onto a point mass. A-LQR (Skifstad et al., 2026) casts steering as linear-quadratic control. Unlike these methods, PTS transports only a k -dimensional projection and leaves its orthogonal complement fixed, so the cost-minimal edit toward a target law is optimal transport in k dimensions, with a closed form for $k = 1$.

Deciding when to act. Activations predict whether an answer is right (Kadavath et al., 2022; Azaria & Mitchell, 2023), and Zhao et al. (2026) trace inflated verbalized confidence to components that write at late token positions. The first generated tokens predict whether steering will succeed (Fan et al., 2026), while probes of a finished behavior forecast it poorly from a prefix (Kortukov et al., 2026). Within one forward pass, a steering write can degrade a direction read after it (Luo et al., 2026). SADI (Wang et al., 2025), dynamic activation composition (Scalena et al., 2024) and GAPS (Fereidouni et al., 2026) adapt the edit per input, per generation step and per dimension, respectively. Closer to our work, three methods separate the decision from the action. CAST (Lee et al., 2025) compares the prompt’s hidden state with a condition vector at a grid-searched threshold and adds a behavior vector when the comparison holds. DSAS (Ferrando et al., 2026) decouples when to steer from how, with a logistic gate that scales any steering map per token. PCHI (Li et al., 2026) conditions a learned rescaling of downstream attention heads on a frozen probe read at a confidence-template token, and we run PTS in its setting (Sec. 6). PTS adds an exact account of the pair (Prop. 1), together with the ceiling and the reference arms it implies, and we run these arms for every conditional method of the multiple-choice comparison.

3 PROJECTION-TRANSPORT STEERING

To steer a behavior only where it occurs, an operator has to settle what edit to apply and to which inputs. PTS answers the first question with a transport map on a behavioral subspace and the second with a likelihood-ratio test on a detection statistic, and it accounts exactly for how the two compose. Proofs are in Appendix B.

The action. Let $h \in \mathbb{R}^d$ be the residual-stream activation at a fixed decoder layer, and let $V \in \mathbb{R}^{k \times d}$ have orthonormal rows spanning what we call the behavioral subspace; $q = Vh$ is the projection. In our experiments $k = 1$, and the row of V is the unit difference of mean activations between confident and hesitant answers, which we call the confidence direction, with q the confidence coordinate; on free text the direction separates toxic from clean continuations. We write μ for the law of h under unsteered generation, μ_V for the induced law of q , and ν for a target law on $\mathbb{R}^k$. Our goal is an edit that gives the projection the law ν , leaves the orthogonal complement untouched, and moves h as little as possible, since large edits degrade generation (Sec. 5).

Definition 1 (admissible steering). $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is admissible for (V, ν) if (i) $(I - V^\top V) T(h) = (I - V^\top V) h$ μ -a.s. (locality: the orthogonal complement is untouched), and (ii) $(V \circ T)_\# \mu = \nu$ (goal: steered projections follow the target law). Its cost is $C(T) = \mathbb{E}_\mu \|T(h) - h\|^2$.

Theorem 1 (steering is optimal transport of the projection). *Let μ_V be absolutely continuous with finite second moment and let ν have a finite second moment. Then $\min_T C(T) = W_2^2(\mu_V, \nu)$ over admissible T , attained by*

$$T^*(h) = h + V^\top (S(Vh) - Vh), \quad (1)$$

where S is the (μ_V -a.s. unique) Brenier map from μ_V to ν . Every optimal T satisfies $VT(h) = S(Vh)$, hence $T = T^*$ by locality, μ -almost surely.

In words, once the subspace and the target law are fixed, the edit is determined: it moves only the projection, by the transfer function S (g in Table 1). For $k = 1$, $S = F_\nu^{-1} \circ F_{\mu_V}$ is the monotone rearrangement, which we store as a 41-knot piecewise-linear lookup; here it suffices that μ_V has no atoms, and ν may have atoms. Directional ablation is then the map onto δ_0 , a clamp at c is the map onto the law of $\min(q, c)$, and the shift $S(q) = q + \alpha$ is the map onto a translate of μ_V ; for Gaussian marginals S is affine (Knott & Smith, 1984; Gelbrich, 1990). Since every nondecreasing transfer function is the optimal map onto its own image law, Theorem 1 turns the choice of edit into a choice of target law, which we make on a tuning split (Sec. 4). For the shift, the theorem gives the following corollary.

Corollary 1 (when additive steering is optimal, and its limit). *(a) $S(q) = q + \alpha$ is the optimal map iff ν is the translate of μ_V by α . (b) Let two subpopulations have the same law of q . If, under an edit, the probability that an input leaves its state depends on h only through q , the edit changes the state of both at the same rate.*

The decision. A decision is needed because the two states we must tell apart look alike to the action. Overconfident-wrong (OCW) and confident-right (CR) answers are both confident, and their laws on the confidence coordinate overlap (Sec. 5), so Corollary 1(b) suggests that a transfer function of that coordinate changes both states at similar rates, as the global shift largely does (Fig. 1). The decision therefore reads a second statistic s of the input and runs the action on an event Γ determined by s , in practice $\{s > \tau\}$ for a scalar score; off Γ the output is exactly the unsteered one. We measure the result with selectivity, $\text{Sel} = P(\text{leave OCW} \mid \text{OCW}) - P(\text{leave CR} \mid \text{CR})$, the share of OCW answers removed minus the share of CR answers lost; on free text, toxic and clean continuations take their places. We write TPR and FPR for the rates at which the decision fires on OCW and CR, and ρ_O, ρ_C for the rates at which the action changes the state of a flagged OCW and a flagged CR answer.

Proposition 1 (selectivity decomposes). *For any decision and action that leave unflagged outputs unchanged,*

$$\text{Sel} = \text{TPR} \cdot \rho_O - \text{FPR} \cdot \rho_C. \quad (2)$$

The proof is one line of total probability. The decision sets (TPR, FPR) and the action sets (ρ_O, ρ_C) on the flagged answers. A gated operator with a post-hoc or prompt decision returns, input by input, either the unsteered or the ungated output, so one ungated run per action composes every such configuration; the ungated rates also predict its selectivity when the action converts flagged and unflagged answers of a state alike (Sec. 5.2).

The identity also fixes three reference arms on the same decision. A no-edit arm measures what regeneration alone changes, and a random direction at the same dose measures what any perturbation of that size changes. An override of the reported confidence of every flagged answer ($\rho_O = \rho_C = 1$; on free text, a refusal) has selectivity $\text{TPR} - \text{FPR}$, and the action is more selective than this override exactly when

$$\text{TPR}(1 - \rho_O) < \text{FPR}(1 - \rho_C), \quad (3)$$

that is, when the share of protected answers flagged but spared exceeds the share of target answers flagged but not converted.

Theorem 2 (optimal decision and selectivity ceiling). *Let p_O, p_C be the densities of s on OCW and CR answers, and $r_O(s), r_C(s)$ the probabilities that the action converts such answers at s . (a) If r_O and r_C do not depend on s , then among decisions with equal FPR the likelihood-ratio test on s maximizes Sel (Neyman–Pearson); under equal-covariance Gaussian class conditionals it is the linear discriminant. In general, among decisions with equal $\text{FPR} \cdot \rho_C$, the maximizer thresholds $p_O(s)r_O(s)/(p_C(s)r_C(s))$. (b) If s is scalar and the concave hull of its ROC has area AUC, every decision $\{s > \tau\}$ has $\text{TPR} - \text{FPR} \leq 2 \text{AUC} - 1$, which caps the override on it and any action with $\rho_O = \rho_C$.*

Part (b) bounds, from unsteered answers alone, what a threshold decision contributes when the action converts both states alike, so the ceiling of a decision is known before any steered rollout; Appendix B treats the general case in which conversion depends on s . In the multiple-choice and free-text experiments the decision is a logistic-regression probe, which estimates the log-likelihood ratio up to a constant that τ absorbs. The probe reads the activations of the finished answer, of its first t tokens or of the prompt, and we fit it on a labeled pool disjoint from every tuning and evaluation split. Once the decision is fixed, Theorem 1 carries over to the inputs it flags.

Theorem 3 (transport under a fixed decision). *Let $P(\Gamma) > 0$, let $\mu_V(\cdot | \Gamma)$ be absolutely continuous, and let ν_Γ have a finite second moment. Among interventions that are local to V , the identity off Γ , and whose projections on Γ follow ν_Γ , the cost-minimal one applies on Γ the Brenier map from $\mu_V(\cdot | \Gamma)$ to ν_Γ , at cost $P(\Gamma) W_2^2(\mu_V(\cdot | \Gamma), \nu_\Gamma)$.*

Proposition 4 replaces the switch by a graded dose, the form of the per-token interpolation of DSAS, and shows that under a linear gain and a price on displacement the threshold gate is optimal exactly when the optimal dose is almost surely 0 or 1. A post-hoc decision scores the finished answer and regenerates the inputs it flags; Proposition 2 states when the decision and the action share one forward pass.

Proposition 2 (single-pass decision). *Let z_t be computed from the activations of the prompt, or of the prompt and the first t generated tokens. Among decisions measurable with respect to z_t , Theorem 2 holds with z_t in place of s , so the ceiling of an early decision is read from the ROC of z_t on unsteered answers. If the action starts after the last position z_t reads, the operator runs in one forward pass: the decision reads only unedited activations, earlier positions keep their cached keys and values, and for $k = 1$ each token costs at most two extra d -dimensional dot products, one lookup and one vector update.*

4 EXPERIMENTAL SETUP

The main model is Gemma-2-2B-it (Gemma Team, 2024) and the main behavior is overconfidence on MMLU (Hendrycks et al., 2021). The multiple-choice comparison is repeated on Qwen2.5-7B-Instruct, a second model family on which every component is refitted, and on ARC-Challenge (Clark et al., 2018). On multiple-choice questions both models reason inside `<think>...</think>` and answer `\boxed{letter}` under greedy decoding. Section 6 adds toxic continuation as a free-text behavior, with Mistral-7B-Instruct-v0.3 as a third family, and the setting of PCHI (Li et al., 2026), the closest conditional method. In both multiple-choice models the action runs at layer 14 (of 26 and 28 layers), where the confidence direction separates confident from hesitant answers of the tuning split within 0.016 AUROC of the best layer (Appendix L).

The MMLU splits are disjoint subsets of its test set: 400 questions to extract the directions and transport targets, a pool of 4,000 to fit the detection probe, 1,200 on which every configuration is tuned, and 3,000 used once for confirmation. On ARC every decision is refitted with 1,012 questions from the ARC training and validation sets added to its pool, 400 further questions from those sets tune the configurations, and the 1,165 test questions confirm them.

The primary readout (M5) scores the answer the model writes. Correctness compares the boxed letter with the gold answer. Confidence is $P(\text{YES})$ when the model is then asked, in the same conversation, whether that answer is correct, and a steered question is steered in both turns. Correctness and whether $P(\text{YES}) \geq 0.5$ define four states; the two confident ones are overconfident-wrong (OCW, wrong) and confident-right (CR, right). Each question is paired with itself before and after the edit. Removal is $P(\text{not OCW} | \text{OCW})$, retention is $P(\text{CR} | \text{CR})$, and Sel = removal $- (1 - \text{retention})$, which is 1 for an edit that removes every OCW and keeps every CR and 0 for one that treats the two states alike. Because selectivity conditions on the starting state, we also report, over all questions, the change in the share of OCW answers, the 10-bin expected calibration error (ECE) of the stated confidence, and the change in accuracy. Appendix E reports two secondary readouts, the letter logit of the same answer and a per-option verification, and Appendix O gives the prompts.

PTS decides post-hoc, from the finished unsteered answer, and then regenerates the flagged questions; at the prompt, in one forward pass over it before generation; or at a prefix of $t \in \{16, 32, 64, 128\}$ generated tokens, which keeps decision and edit in one pass. Its action moves only the projection. On multiple-choice questions it is a shift at one of several doses or an ablation,

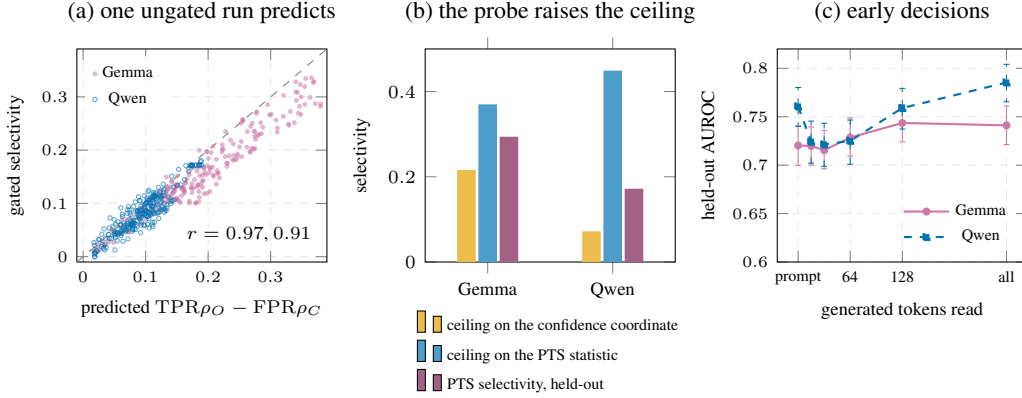


Figure 2: Proposition 1 predicts gated selectivity from ungated runs, and the PTS probe raises the ceiling of a threshold decision. (a) Gated selectivity of 480 tuning-split configurations (six actions, two decisions and 20 thresholds per model) against its prediction, with equality dashed; (b) ceiling $\max(\text{TPR} - \text{FPR})$ of a threshold on each statistic, from unsteered held-out answers, next to held-out PTS selectivity; (c) held-out AUROC of the PTS probe when it reads the prompt, a prefix of the answer or the whole answer.

both transport maps of Theorem 1; on free text it can also be the displacement of the quantile map onto the projection law of clean continuations, scaled by $\lambda \in \{1, 2, 3, 4, 6\}$, which at $\lambda = 1$ is the map itself.

We compare PTS with seven baselines: prompting with a calibration instruction, additive CAA (Panickssery et al., 2024), directional ablation (Arditi et al., 2024), MiMiC (Singh et al., 2024), Linear-AcT (Rodriguez et al., 2025a), CAST with its prompt condition (Lee et al., 2025) fitted on the same labeled pool as our probe, and a CAST-style trace condition, a shift conditioned on the finished trace that, like post-hoc PTS, regenerates the questions it flags. On free text the baselines are prompting, the global shift and CAST. All methods are reimplemented on our hooks and scored with the same readout. One rule selects every configuration on the tuning split: the highest selectivity among configurations whose accuracy change is at least -0.01 , or the highest overall when none meets that floor; every confirmatory number is a real run. The budgets are one configuration for prompting and for ablation, eight for CAA, nine each for MiMiC and Linear-AcT, and sixteen for each conditional method, ours and the two CAST variants alike (Appendix C).

Every conditional arm is run against three references that share its decision: a no-edit arm, which regenerates the flagged inputs with the identity action; random unit directions at the same dose (ten per conditional multiple-choice arm; one hundred for PTS, or thirty displaced as far as the ablation selected on the second family); and an override that marks every flagged answer as not confident or, on free text, replaces every flagged continuation by a refusal. Intervals are 95% paired bootstrap intervals over questions (prompts on free text) with 10^4 resamples. The seven comparisons of PTS with the baselines are two-sided and Holm-adjusted within each setting, each later protocol Holm-adjusts its own tests, and an accuracy change is non-inferior when the lower end of its interval is above -0.02 . Every protocol was committed before its held-out rollouts, the multiple-choice one with the hashes of its selected configurations and fitted detectors (Appendix A).

5 SELECTIVE REMOVAL OF OVERCONFIDENT ANSWERS

The PTS probe raises the selectivity ceiling of a threshold decision (Theorem 2b) above that of the confidence coordinate on both models. On the held-out questions the two confident states overlap on the coordinate, with AUROC 0.635 on Gemma and 0.53 on Qwen, in line with the frontier of the global shift in Fig. 1a. The best threshold on the coordinate separates them by $\max(\text{TPR} - \text{FPR}) = 0.216$ and 0.072 . With the PTS probe, fitted on a separate pool of 4,000 labeled answers, the ceiling is 0.370 and 0.449 (AUROC 0.741 and 0.785; Fig. 2b).

method	Δacc	removal	retention	Sel $\uparrow$ [95% CI]	ΔSel (p)	ΔOCW (pp) $\downarrow$	ΔECE $\downarrow$
prompting	+0.005	0.42	0.78	0.200 [0.16, 0.24]	-0.094 (<0.001)	-5.9*	-0.030*
additive CAA	-0.001	0.56	0.66	0.220 [0.18, 0.26]	-0.074 (0.008)	-13.0*	-0.153*
dir. ablation	+0.009	0.46	0.72	0.173 [0.13, 0.21]	-0.121 (<0.001)	-8.7*	-0.111*
MiMiC	+0.000	0.31	0.80	0.117 [0.08, 0.15]	-0.177 (<0.001)	-2.1*	-0.018*
Linear-AcT	+0.005	0.14	0.92	0.061 [0.04, 0.09]	-0.233 (<0.001)	-0.2	-0.003
CAST (prompt)	-0.002	0.46	0.76	0.217 [0.18, 0.26]	-0.077 (0.004)	-11.2*	-0.137*
CAST-style (trace)	-0.005	0.54	0.70	0.248 [0.21, 0.29]	-0.046 (0.020)	-16.6*	-0.094*
PTS, post-hoc	+0.003	0.49	0.80	0.294 [0.26, 0.33]	—	-15.0*	-0.099*
PTS, prefix	-0.008	0.56	0.73	0.294 [0.26, 0.33]	+0.000	-17.2*	-0.136*
PTS, prompt	-0.012 [†]	0.55	0.76	0.306 [0.27, 0.35]	+0.012	-17.6*	-0.084*
same decision, no edit	+0.003	—	—	0.049 [0.03, 0.07]	-0.245 (<0.001)	—	-0.005
same decision, random direction	—	—	—	0.131	-0.163 (<0.001)	—	—
same decision, override readout	+0.000	—	—	0.319 [0.28, 0.36]	+0.025	—	—

Table 2: PTS is the most selective of eight methods on held-out questions. Gemma-2-2B, 3,000 MMLU questions (unsteered accuracy 0.559, ECE 0.354; $n_{OCW}=963$, $n_{CR}=1,399$), each method at the configuration selected on the tuning split; ΔSel is the row minus PTS post-hoc, with Holm-adjusted p over the seven baselines. The middle block takes the PTS decision before the answer is finished; the last three rows keep the PTS decision with no edit, with a random direction at the PTS dose (mean of 100), and with the reported confidence overridden (Eq. (3)). *: 95% interval excludes 0; [†]: accuracy interval reaches -0.02.

5.1 PTS ON HELD-OUT QUESTIONS

Post-hoc PTS is the most selective of the eight methods on the 3,000 held-out questions (Table 2). It removes 49% of overconfident-wrong answers and keeps 80% of confident-right ones, a selectivity of 0.294 [0.256, 0.334]. The closest baseline, the CAST-style trace condition, reaches 0.248, and the other six trail PTS by 0.074 to 0.233; all seven differences survive Holm adjustment. The lead over the CAST-style condition also holds when the confidence threshold that defines the states moves from 0.5 to 0.4 (+0.044 [0.007, 0.080]) or to 0.6 (+0.068 [0.028, 0.107]).

Over all questions, PTS lowers the share of overconfident-wrong answers by 15.0 points [13.7, 16.4] and the calibration error from 0.354 to 0.255, with a non-inferior accuracy change of +0.003 [-0.009, 0.015]. The decision can also be taken before the answer is finished. The probe separates the two states from the prompt almost as well as from the finished answer (Fig. 2c), and a decision read from the first 32 generated tokens, in the same pass as the edit, gives the same selectivity, 0.294.

The edit along the confidence direction carries this selectivity. Under the PTS decision, regenerating the flagged questions without an edit gives 0.049, and one hundred random unit directions at the PTS dose average 0.131; PTS lies above all hundred and above their mean by 0.163 [0.140, 0.188] (gray points in Fig. 1a). On code synthesis the prompt decision keeps the edit off: it fires on none of the 164 HumanEval problems, so pass@1 stays at its unsteered 0.396, whereas the ungated shift at its selected dose lowers it to 0.281 (Appendix M).

5.2 GATED SELECTIVITY FROM ONE UNGATED RUN

Proposition 1 predicts the selectivity of every decision placed in front of an action from one ungated run of that action. On the tuning split we measured the conversion rates of six actions once, composed each with two decisions at 20 thresholds, and compared the prediction $\text{TPR } \rho_O - \text{FPR } \rho_C$ with the gated selectivity composed from the same run (Fig. 2a). The correlation is 0.97 over 240 configurations on Gemma and 0.91 over 240 on Qwen. The composition that scores gated configurations during selection also matches real gated runs: on the three held-out splits, the twelve real gated arms lie within 0.025 of their composition, with a median gap of 0.005. For post-hoc and prompt decisions the tuning search thus needs one ungated generation per action (Appendix D).

6 PTS BEYOND THE MAIN SETTING

PTS stays selective on a second model family and on a second benchmark (Table 3). The tuning rule selects ablation as the action on Qwen, where PTS reaches selectivity 0.172 [0.136, 0.210], 0.081 above regenerating the flagged questions without an edit and 0.132 above the CAST-style trace condition. On ARC-Challenge PTS reaches 0.235 [0.168, 0.303] and lowers the calibration error by 0.060. There it is above Linear-AcT and the CAST prompt condition after Holm adjustment, and above the 95th percentile of a hundred random directions sent through its decision.

method	Qwen2.5-7B, MMLU ($n_{OCW}=563$, $n_{CR}=2,007$)				Gemma-2-2B, ARC ($n_{OCW}=228$, $n_{CR}=780$)			
	Δ_{acc}	Sel $\uparrow$ [95% CI]	ΔSel (p)	$\Delta ECE \downarrow$	Δ_{acc}	Sel $\uparrow$ [95% CI]	ΔSel (p)	$\Delta ECE \downarrow$
prompting	-0.003	0.212 [0.17, 0.25]	+0.040 (0.313)	-0.004	+0.003	0.238 [0.17, 0.31]	+0.003 (1.000)	+0.009
additive CAA	-0.015 [†]	0.151 [0.12, 0.19]	-0.021 (0.998)	+0.012	-0.001 [†]	0.294 [0.22, 0.37]	+0.059 (0.356)	-0.069*
dir. ablation	-0.009 [†]	0.156 [0.12, 0.19]	-0.016 (0.998)	+0.005	-0.008 [†]	0.215 [0.14, 0.29]	-0.020 (1.000)	-0.055*
MiMiC	-0.002	0.154 [0.12, 0.19]	-0.018 (0.998)	+0.020*	+0.010	0.198 [0.13, 0.26]	-0.037 (1.000)	-0.007
Linear-AcT	-0.003	0.123 [0.09, 0.16]	-0.049 (0.133)	+0.009	+0.003	0.109 [0.06, 0.16]	-0.126 (0.008)	+0.000
CAST (prompt)	-0.011 [†]	0.122 [0.09, 0.16]	-0.050 (0.133)	+0.027*	-0.003	0.109 [0.05, 0.17]	-0.126 (0.003)	-0.028*
CAST-style (trace)	-0.007	0.040 [0.02, 0.06]	-0.132 (<0.001)	+0.013*	+0.001	0.289 [0.22, 0.36]	+0.054 (0.356)	-0.061*
PTS, post-hoc	-0.002	0.172 [0.14, 0.21]	—	-0.000	-0.006 [†]	0.235 [0.17, 0.30]	—	-0.060*
PTS, prefix	-0.008	0.109 [0.08, 0.14]	-0.063	+0.013*	-0.008 [†]	0.276 [0.21, 0.35]	+0.041	+0.000
PTS, prompt	-0.034 [†]	0.120 [0.08, 0.16]	-0.052	+0.049*	-0.027 [†]	0.182 [0.11, 0.25]	-0.053	+0.028*
same decision, no edit	-0.002	0.091 [0.06, 0.12]	-0.081 (<0.001)	+0.000	-0.004	0.016 [-0.01, 0.05]	-0.219 (<0.001)	+0.011
same decision, random direction	—	0.158	-0.014 (0.175)	—	—	0.167	-0.068 (0.004)	—
same decision, override readout	+0.000	0.391 [0.36, 0.43]	+0.218	—	+0.000	0.270 [0.20, 0.34]	+0.035	—

Table 3: PTS is above its no-edit reference arm on a second model family (left) and on a second benchmark (right). Notation as in Table 2; unsteered accuracy is 0.737 on Qwen and 0.744 on ARC. The random-direction row averages 100 directions on ARC and, on Qwen, 30 directions displaced as far as the selected ablation; the last row overrides the reported confidence of the answers the PTS decision flags.

method	Gemma-2-2B (toxic 15.0%)			Qwen2.5-7B (toxic 20.3%)			Mistral-7B (toxic 3.4%)		
	rem./kept	Sel $\uparrow$ [95% CI]	ΔSel (p)	rem./kept	Sel $\uparrow$ [95% CI]	ΔSel (p)	rem./kept	Sel $\uparrow$ [95% CI]	ΔSel (p)
prompting	42/68	0.104 [0.06, 0.15]	-0.422 (<0.001)	59/79	0.386 [0.34, 0.43]	-0.148 (<0.001)	72/70	0.419 [0.33, 0.51]	-0.101 (0.084)
global shift	90/58	0.474 [0.44, 0.51]	-0.051 (0.010)	84/57	0.403 [0.37, 0.44]	-0.131 (<0.001)	71/72	0.429 [0.34, 0.52]	-0.091 (0.134)
CAST	81/70	0.515 [0.47, 0.56]	-0.011 (0.585)	74/70	0.443 [0.40, 0.48]	-0.091 (<0.001)	64/81	0.446 [0.35, 0.54]	-0.074 (0.220)
PTS, shift	73/80	0.526 [0.48, 0.57]	—	82/72	0.534 [0.50, 0.57]	—	74/78	0.520 [0.43, 0.61]	—
PTS, scaled transport	79/79	0.584 [0.54, 0.62]	+0.058 (<0.001)	—	—	—	—	—	—
PTS, post-hoc	91/81	0.723 [0.69, 0.75]	+0.198 (<0.001)	76/89	0.648 [0.61, 0.68]	+0.114 (<0.001)	88/83	0.712 [0.65, 0.77]	+0.192 (<0.001)
same decision, random direction	—	0.099	-0.427 (<0.001)	—	0.184	-0.349 (<0.001)	—	0.247	-0.273 (<0.001)
same decision, refusal	82/60	0.427	-0.099 (<0.001)	86/59	0.457	-0.077 (<0.001)	81/49	0.305	-0.214 (<0.001)

Table 4: Among single-pass methods, PTS is the most selective on free text in all three families. 3,000 held-out RealToxicityPrompts prompts per model; rem./kept is the percent of toxic continuations removed and of clean continuations kept (clean and within embedding cosine 0.5 of the unsteered text), and ΔSel is the row minus PTS with the shift. Post-hoc PTS regenerates the prompts it flags, the scaled transport displacement is the tuning-rule choice on Gemma, and the last two rows keep the PTS decision.

6.1 THE SETTING OF PCHI

In the setting of PCHI (Li et al., 2026), PTS is more selective than the learned head rescaling under the same probe. PCHI gates a rescaling of 128 attention heads with a probe read where the model states its confidence. We rebuild this setting: Qwen3-4B-Instruct answers OpenMathInstruct-2 problems in a JSON format that ends with a yes/no confidence field, and a probe on the layer-18 state at that field flags overconfident-wrong answers. PTS keeps the probe and replaces the learned rescaling with a shift of the same state along one direction. On 3,183 held-out answers it reaches selectivity 0.438, against 0.354 for PCHI (+0.084 [0.058, 0.110]), and a calibration error of 0.188 against 0.234 (Appendix J).

6.2 FREE TEXT: TOXIC CONTINUATION

We test the action on free text, where a continuation has no single readout that a decision could overwrite. We steer toxic continuation of RealToxicityPrompts (Gehman et al., 2020), the setting in which Linear-AcT reports toxicity mitigation, in all three model families (Appendix H). A RoBERTa classifier labels a continuation toxic at score 0.5. A clean continuation counts as kept only if the steered one is clean and within sentence-embedding cosine 0.5 of the unsteered one, so rewriting a clean text counts as damage. The direction is the difference of mean activations between toxic and clean continuations, and the decision is a probe on the prompt, read before generation. Because the decision gates each row inside its original batch, every continuation it does not flag is identical to the unsteered one.

PTS is more selective than refusal on its own decision in every family, and it lies above the 95th percentile of a hundred random directions sent through that decision (Table 4). For every gated arm that we compared with refusal, Eq. (3) evaluated with tuning-split conversion rates predicts the held-out winner. On Gemma and Qwen, PTS is also more selective than the global shift and prompting, and on Qwen it is above CAST, 0.534 against 0.443. With Mistral, whose continuations are rarely toxic, PTS lowers the toxic rate from 3.4% to 1.4% at selectivity 0.520 [0.430, 0.605].

On Gemma the tuning rule selects an edit along the transport displacement, and this edit is more selective than a shift of the same mean displacement. The edit is $h + 3V^\top(S(Vh) - Vh)$, where S is the quantile map from the law of the projection over toxic continuations to its law over clean

ones. It reaches selectivity 0.584 [0.542, 0.624] and lowers the toxic rate from 15.0% to 3.5% with 79% of clean continuations kept. Its lead is 0.043 [0.012, 0.076] over the matched shift and 0.069 [0.021, 0.117] over CAST.

On a second, previously unused set of 3,000 prompts per model, under a protocol pushed before any steered rollout, PTS again exceeds refusal on its own decision by 0.15 to 0.17 and a hundred random directions by 0.23 to 0.43 in every family. It is also above prompting in all three (Appendix I).

6.3 INFERENCE COST

With a prompt decision, PTS adds 9% to generation time. On one H100, 750 toxicity continuations of the main model take 5.79 seconds unsteered and 6.31 seconds with PTS, of which 0.46 seconds go to the forward pass over the prompts that reads the decision. A post-hoc decision regenerates the flagged half and takes 8.74 seconds (Appendix K).

7 LIMITATIONS

Overriding the single readout of multiple-choice confidence on every answer the post-hoc PTS decision flags is at least as selective as the action (Appendix G). On a fresh free-text split, PTS does not exceed a per-token gate without the input decision, and the monotone transport map is level with an equal-displacement shift (Appendix I). The evidence covers two behaviors and three families of 2B to 7B models with one toxicity classifier; the multiple-choice protocols became public with their results.

8 CONCLUSION

We have introduced PTS, which splits an activation edit into an action, a transport map on the projection onto a behavioral subspace, and a decision, a likelihood-ratio test that sets where the action runs. Once the subspace and the target law are fixed, the cost-minimal action is the optimal-transport map between the projection laws. The selectivity of the composed operator equals the decision’s error rates weighted by the action’s conversion rates, so one ungated run predicts gated configurations and unsteered answers alone give the ceiling of a threshold decision. In the main multiple-choice setting PTS is the most selective of eight methods on held-out questions, and on free-text toxicity it is more selective than refusal on its own decision in all three model families.

AI USE STATEMENT

We used generative AI tools to polish the manuscript and to retrieve related work. The authors verified every resulting claim and reference and take responsibility for the final content.

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A CONFIRMATORY PROTOCOL

The multiple-choice protocols (v19 for MMLU, v20 for ARC) were committed to the local repository before any steered rollout on their confirmatory splits was generated and first pushed together with the results, so their ordering rests on local commit times; the later protocols v21 to v24 were pushed to a public repository, and so timestamped by the hosting service, before their rollouts existed (`rebuttal/protocols/confirmatory-v19.md`, with two amendments made after the tuning pass and before launch), together with the SHA-256 hashes of the selected configurations and of every fitted detector (`confirmatory-v19-gemma.sha256`).

Splits. All splits are subsets of the 14,042-question MMLU test set, identified by index and constructed by `label_pool.split_records`. The extraction split (400) excludes the first evaluation subset of the pilot protocol; the tuning split (1,200) excludes the extraction split and the five pilot evaluation subsets; the detector pool (4,000) and the confirmatory split (3,000) are consecutive blocks of a seeded permutation of the remaining questions. No question appears in two of the four splits.

Selection rule. For every method the configuration with the highest tuning-split selectivity among those with $\Delta_{\text{acc}} \geq -0.01$ is carried unchanged to the confirmatory split. Gated configurations are scored on the tuning split by combining the ungated rollouts of the same split with the gate’s flags (Prop. 1 makes this exact up to batched-decoding nondeterminism; Appendix D); single-pass configurations are real runs. Every confirmatory number is a real run.

Tests. Primary: PTS post-hoc against each of the seven baselines on pooled M5 selectivity, question-level paired bootstrap with 10^4 resamples, two-sided, Holm-adjusted over the seven comparisons. Accuracy: an arm is non-inferior if the lower end of the 95% interval of its accuracy change is above -0.02 . The single-pass and prompt-decision variants, M4 and M2 selectivities, removal, retention and ECE are secondary and reported without a claim.

Released records. Every unsteered and steered answer of the tuning and confirmatory splits is released with its state, confidence and correctness (`results/release/records.csv.gz`); `paper/recompute_from_records.py` recomputes each selectivity and paired interval in Tables 2 and 3, including the null arms, from that file alone. `results/release/toxicity.csv.gz` holds every confirmatory continuation of the free-text setting with its judge score and state (texts omitted for the random-direction arms), and `results/release/pchi_val.csv.gz` the held-out readout gaps of every arm of the PCHI setting.

ARC replication. A second protocol (`confirmatory-v20-arc.md`), frozen before any steered ARC rollout, repeats the Gemma comparison on ARC-Challenge: a detector pool of 1,012 and a tuning split of 400 from the ARC training and validation questions, and the 1,165-question test split as the confirmatory split. The directions, moments and action layer are those fitted on MMLU; every decision is refitted on the ARC pool together with the MMLU pool, and every configuration is selected on the ARC tuning split with the same rule and grids.

Null arms. A third protocol (`confirmatory-v21-null.md`), with the job list and the hash of the runner, was committed and pushed to a public repository before any null rollout existed; the commit is timestamped by the hosting service. On each confirmatory split it adds the frozen PTS decision with an identity action, the frozen decision with the selected action on a unit random direction (seed 0), and an ungated rerun with an identity action. It registers two one-sided hypotheses per setting, PTS above each gated null, with Holm adjustment over the six tests, and the net measures of Appendix F for every arm. Five of the six tests pass; PTS against the single random direction on ARC gives $+0.041 [-0.029, 0.109]$, Holm $p = 0.12$.

Reference arms. A fourth protocol (`confirmatory-v22-references.md`), pushed before any of its rollouts existed, extends the no-edit null to all four conditional methods, replaces the single random direction by ten (seeds 0 to 9), and adds random directions for the sycophancy edit. It registers 24 one-sided tests (four methods, three settings, no-edit and random-edit null), Holm-adjusted together, of which 18 pass (Table 13); a count of the cases in which Eq. (3) with ungated conversion rates predicts the sign of steer minus matched override (10 of 13); and paired intervals for selectivity at cutoffs 0.4 and 0.6 and over boxed traces for PTS against the CAST-style condition on Gemma. The v22 random-edit null supersedes the single-direction test of v21.

Free text and dense nulls. A fifth protocol (`confirmatory-v23-freetext.md`), pushed with the hashes of the tuning selections, fitted decisions and code before any confirmatory rollout of the toxicity setting existed, fixes the data, judge, retention criterion, arms and ten primary hypotheses of Appendix H, and extends the random directions of PTS post-hoc to one hundred on Gemma-MMLU and Gemma-ARC and to thirty displacement-matched directions on Qwen (Table 14).

Transport map and a third family. A seventh protocol (`confirmatory-v25-transport-mistral.md`), pushed before its rollouts, adds the quantile transport map to the action family on free text, records its selection on the Gemma tuning split, and registers five tests: the transport map against a shift of equal mean displacement on Gemma, and PTS against the global shift, prompting, refusal on its decision and a hundred random directions on Mistral-7B. Three of the five pass after Holm adjustment, including the transport-map test ($p = 0.003$).

PCHI setting. A sixth protocol (`confirmatory-v24-pchi.md`), pushed after the train-split fitting and tuning and before any validation rollout, fixes the arms and three hypotheses of Appendix J.

Analyses added after the protocol. The detect-and-override reference, the difference-of-means decision fitted on 4,400 labels, the latency measurement (Appendix K), the layer analysis (Appendix L), calibration and the out-of-domain negatives were run after the confirmatory results were known. They use the same splits and the frozen decisions, and we report them as exploratory.

B PROOFS

Theorem 1. By locality, $T(h) - h \in \text{span}(V^\top) \mu\text{-a.s.}$, hence $\|T(h) - h\|^2 = \|VT(h) - Vh\|^2$ because the rows of V are orthonormal, and $C(T) = \mathbb{E}\|VT(h) - Vh\|^2$. The joint law γ of $(Vh, VT(h))$ is a coupling of (μ_V, ν) by the goal constraint, so $C(T) = \int \|y - x\|^2 d\gamma \geq$

$W_2^2(\mu_V, \nu)$. Because μ_V is absolutely continuous and both laws have finite second moments, Brenier's theorem gives a unique optimal coupling, induced by $S = \nabla\varphi$ with φ convex. $T^*(h) = h + V^\top(S(Vh) - Vh)$ is admissible ($VV^\top = I_k$ gives $VT^*(h) = S(Vh)$, and the complement is untouched) and attains the bound. For any minimiser the induced coupling is optimal, hence equal to the unique one, so $VT(h) = S(Vh)$ a.s. $\square$

For $k = 1$ absolute continuity can be weakened to atomlessness of μ_V : the monotone rearrangement $F_V^{-1} \circ F_{\mu_V}$ is then an optimal map for every ν with a finite second moment, including targets with atoms such as δ_0 (ablation) or a law truncated at c (clamp); uniqueness of the map can fail only on a μ_V -null set.

Proposition 3 (metric covariance). *If the cost is $(\Delta h)^\top M(\Delta h)$ with $M \succ 0$, the admissible problem is optimal transport of μ_V to ν in $\mathbb{R}^k$ under the cost $\|\cdot\|_{M_V}^2$, $M_V = VMV^\top \succ 0$, and the optimal operator is $T^*(h) = h + V^\top(S_M(Vh) - Vh)$ with $S_M = M_V^{-1/2} \circ \nabla\varphi \circ M_V^{1/2}$, φ convex.*

Proof. Locality writes $\Delta h = V^\top a$ with $a \in \mathbb{R}^k$, and the goal constraint with $VV^\top = I_k$ fixes $a = S(Vh) - Vh$ for the map S in use. Then $\|\Delta h\|_M^2 = a^\top M_V a$. The cost is a latent-space Mahalanobis cost on $\mathbb{R}^k$; it is not the full-space Mahalanobis distance of Δh unless M commutes with $V^\top V$. In the coordinates $z = M_V^{1/2} q$ it is Euclidean, the change of variables is internal to $\mathbb{R}^k$, and Brenier's theorem in z (the pushforward of an absolutely continuous law under an invertible linear map is absolutely continuous) gives $\nabla\varphi$; mapping back gives S_M . $\square$

Corollary 1. (a) If $\nu = \mu_V * \delta_\alpha$ the map $x \mapsto x + \alpha$ is the gradient of a convex function pushing μ_V to ν , hence the Brenier map. Conversely, if the optimal map is $x \mapsto x + \alpha$ then ν is its pushforward, the shift. For one-dimensional Gaussians the monotone map is $x \mapsto m_t + (\sigma_t/\sigma_s)(x - m_s)$, with slope σ_t/σ_s . (b) If two groups of inputs have the same law of p and the probability that an input changes state depends on h only through p , every transfer function changes both groups at the same rate, since both rates are the same integral against the law of p . A decision that reads p alone therefore adds little, which is why the detection statistic reads the full activation. $\square$

Proposition 1. Let R be the event that an OCW trace leaves OCW and L the event that a CR trace leaves CR. Off Γ the output equals the unsteered output, which is in its original state by definition, so $R \subseteq \Gamma$ and $L \subseteq \Gamma$ up to null sets. Then $P(R \mid OCW) = P(\Gamma \mid OCW)P(R \mid OCW, \Gamma) = \text{TPR} \rho_O$, and likewise $P(L \mid CR) = \text{FPR} \rho_C$; subtracting gives Eq. (2). If, within OCW, the indicator that the action converts the trace is independent of Γ , then $\rho_O = P(R^{\text{ungated}} \mid OCW)$, and likewise for CR; the ungated selectivity is $\rho_O - \rho_C$, and the difference is $\rho_C(1 - \text{FPR}) - \rho_O(1 - \text{TPR})$. $\square$

The identity requires that the unflagged output be exactly the unsteered one. The post-hoc gate satisfies this by construction. The single-pass gate satisfies it for traces on which it never fires, since the action hook is the identity until the decision token.

Theorem 2. Write p_O, p_C for the densities of s on OCW and CR answers and $r_O(s), r_C(s)$ for the probabilities that the action converts an answer of each state with statistic s . For a decision region Γ , $\text{TPR} \rho_O = \int_\Gamma p_O r_O$ and $\text{FPR} \rho_C = \int_\Gamma p_C r_C$, so $\text{Sel}(\Gamma) = \int_\Gamma (p_O r_O - p_C r_C)$. Among regions with a given value of $\int_\Gamma p_C r_C$, the Neyman-Pearson lemma applied to the nonnegative measures $p_O r_O$ and $p_C r_C$ shows that $\int_\Gamma p_O r_O$ is maximised by a superlevel set of their ratio, which is the general statement in (a). If r_O and r_C are constant in s , the ratio is a constant multiple of p_O/p_C , so every objective increasing in TPR and decreasing in FPR is maximised on the likelihood-ratio ROC curve, which is (a). The assumption matters: with $s \in \{a, b\}$, $p_O(a) = p_O(b) = \frac{1}{2}$, $p_C(a) = \frac{1}{4}$, $p_C(b) = \frac{3}{4}$, $r_O(a) = 0.2$, $r_O(b) = 1$, $r_C(a) = 1$ and $r_C(b) = 0$, the likelihood ratio ranks a above b , yet $\Gamma = \{b\}$ gives selectivity 0.5, against -0.15 for $\{a\}$ and 0.35 for $\{a, b\}$. On our data r does depend on s : flagged answers are converted more often than unflagged answers of the same state (Appendix D). Our detector is a logistic-regression probe, a discriminative estimate of the log-likelihood ratio up to the prior term; (a) states what the probe approximates, not that the fitted probe is optimal. (b) The concavified ROC contains the path $(0, 0) \rightarrow (\text{FPR}^*, \text{TPR}^*) \rightarrow (1, 1)$ through the Youden-optimal point, whose area is $\frac{1}{2}(1 + J)$, so $J \leq 2 \text{AUC} - 1$. The override reference has selectivity $\text{TPR} - \text{FPR} \leq J$, and with $\rho_O = \rho_C = \rho$, Eq. (2) gives $\text{Sel} = \rho(\text{TPR} - \text{FPR}) \leq J$. $\square$

Theorem 3. Split $C(T)$ over Γ and Γ^c . The second term is zero by the identity constraint; the first, with the conditional goal constraint, is Theorem 1 for $\mu(\cdot \mid \Gamma)$, which applies when $\mu_V(\cdot \mid \Gamma)$ is absolutely continuous. $\square$

This result fixes the decision. It does not say that a threshold gate is the best decision for the displacement-priced objective; the next statement shows that it generally is not, and when it is.

Proposition 4 (graded dose). Let $T_\lambda(h) = h + \lambda(s) V^\top (S(Vh) - Vh)$ with a dose $\lambda : \mathbb{R} \rightarrow [0, 1]$ measurable in the detector statistic s . Write $\pi(s) = P(OCW \mid s)$, $d(s) = \mathbb{E}[\|S(Vh) - Vh\|^2 \mid s] > 0$, and suppose the behavioral gain is linear in the dose, with slope $\beta_1 > 0$ on OCW and $-\beta_0 < 0$ on CR . The dose maximising gain minus displacement priced at $1/(2c)$ is

$$\lambda^*(s) = \left[\frac{c(\beta_0 + \beta_1)(\pi(s) - \tau)}{d(s)} \right]_0^1, \quad \tau = \frac{\beta_0}{\beta_0 + \beta_1}.$$

It is zero exactly on $\{\pi \leq \tau\}$ and nondecreasing in the ratio $(\pi(s) - \tau)/d(s)$. It is nondecreasing in π when d is constant, but not in general. The threshold gate $\mathbf{1}[\pi > \tau]$ is optimal iff $\lambda^* \in \{0, 1\}$ almost surely, and strictly suboptimal iff $P(0 < \lambda^* < 1) > 0$.

Proof. The displacement of T_λ is $\lambda(s)^2 \|S(Vh) - Vh\|^2$, with expectation $\mathbb{E}_s[\lambda(s)^2 d(s)]$; the gain is $\mathbb{E}_s[\lambda(s)((\beta_0 + \beta_1)\pi(s) - \beta_0)]$. The objective is an integral over s of a strictly concave quadratic in λ , maximised pointwise on $[0, 1]$ by the clipped stationary point. It is the unique maximiser a.s., so the indicator is optimal iff it agrees with λ^* a.s., i.e. iff $\lambda^* \in \{0, 1\}$ a.s.; otherwise the objective is strictly larger at λ^* on a set of positive measure. For non-monotonicity in π , take s_1, s_2 with $\pi(s_1) = 0.6 < \pi(s_2) = 0.7$, $\tau = 0.5$, $c(\beta_0 + \beta_1) = 1$, $d(s_1) = 0.2$ and $d(s_2) = 1$: then $\lambda^*(s_1) = 0.5 > \lambda^*(s_2) = 0.2$. For a counterexample to strict suboptimality at every finite price, let π take only the values 0.1 and 0.9 with $\tau = 0.5$ and $d \leq 0.4c(\beta_0 + \beta_1)$: then $\lambda^* \in \{0, 1\}$ a.s. and the threshold gate is optimal. $\square$

Proposition 2. Let $\mathcal{F}_t$ be generated by the first t generated tokens and z_t the mean detector feature over them. A decision measurable with respect to $\sigma(z_t)$ is a test between the class-conditional laws of z_t , so Theorem 2 applies with z_t in place of s and nothing is assumed about how z_t relates to the finished trace. The reader hook sits at a layer at or above the actor; within one forward pass the actor at position t sees the decision accumulated over positions $< t$, the running mean needs one sum and one counter per sequence, and an edit at position t changes only the keys and values written at t .

Proposition 5 (computational cost). Any algorithm implementing a non-degenerate intervention local to a subspace spanned by a dense direction reads all d coordinates of h , so it costs $\Omega(d)$ per token. The edit of Eq. (1) costs $\Theta(kd + k \log K)$ with a K -knot lookup, and the detector adds one d -dimensional dot product per token.

Proof. If a coordinate i with $(\hat{v})_i \neq 0$ is never read, an adversary perturbs h_i , which changes $p = \langle h, \hat{v} \rangle$ and, on a set of positive measure, the correct output, while the algorithm’s output is unchanged. The upper bound counts Vh , a binary search in the lookup, and $h + V^\top(\cdot)$. $\square$

C TUNING GRIDS

Every configuration searched on the tuning split, with the selected one marked $\star$. Gated rows are scored by composing the ungated rollouts of the same split with the decision’s flags (Appendix D); single-pass rows are real runs. The rule is the highest selectivity among configurations with $\Delta_{\text{acc}} \geq -0.01$, and, when no configuration of a family meets the floor, the highest selectivity overall; this fallback applies to the additive, CAST and prompt-decision families on Qwen, where every shift in the grid costs more than one accuracy point.

Table 5: Tuning split, gemma-2-2b-it ($n = 1,200$, M5).

family	configuration	Δ_{acc}	removal	retention	Sel
additive CAA	alpha-0.375	-0.022	0.831	0.424	+0.255
additive CAA	alpha-0.25 $\star$	-0.002	0.536	0.661	+0.197
additive CAA	alpha-0.5	-0.042	0.974	0.136	+0.110
additive CAA	alpha-0.125	+0.010	0.315	0.763	+0.078
additive CAA	alpha-0.75	-0.153	1.000	0.000	+0.000
additive CAA	alpha-1.0	-0.205	1.000	0.000	+0.000
additive CAA	alpha-1.25	-0.270	1.000	0.000	+0.000

	family	configuration	Δ acc	removal	retention	Sel
756	additive CAA	alpha-1.5	-0.299	1.000	0.000	+0.000
757	MiMiC	L10.reg0.0001★	+0.022	0.349	0.823	+0.172
758	MiMiC	L10.reg0.1	+0.012	0.349	0.809	+0.158
759	MiMiC	L10.reg0.01	+0.007	0.320	0.814	+0.135
760	MiMiC	L18.reg0.1	+0.022	0.260	0.860	+0.121
761	MiMiC	L18.reg0.01	+0.026	0.242	0.869	+0.111
762	MiMiC	L18.reg0.0001	+0.023	0.250	0.857	+0.107
763	MiMiC	L14.reg0.01	+0.001	0.224	0.857	+0.081
764	MiMiC	L14.reg0.1	+0.004	0.216	0.850	+0.066
765	MiMiC	L14.reg0.0001	-0.013	0.214	0.841	+0.055
766	Linear-AcT	L10.lam0.5★	+0.016	0.133	0.936	+0.069
767	Linear-AcT	L10.lam1.0	+0.015	0.138	0.919	+0.057
768	Linear-AcT	L18.lam1.0	+0.012	0.130	0.926	+0.056
769	Linear-AcT	L14.lam0.25	+0.007	0.128	0.924	+0.052
770	Linear-AcT	L14.lam1.0	+0.009	0.138	0.913	+0.051
771	Linear-AcT	L14.lam0.5	+0.010	0.125	0.919	+0.044
772	Linear-AcT	L18.lam0.5	+0.004	0.117	0.926	+0.043
773	Linear-AcT	L10.lam0.25	+0.010	0.112	0.919	+0.031
774	Linear-AcT	L18.lam0.25	+0.005	0.096	0.922	+0.019
775	CAST-style, trace condition	tau.cr.q30.alpha-0.375	-0.020	0.815	0.458	+0.273
776	CAST-style, trace condition	tau.cr.q50.alpha-0.375	-0.015	0.732	0.532	+0.264
777	CAST-style, trace condition	tau.cr.q70.alpha-0.5	-0.014	0.638	0.611	+0.249
778	CAST-style, trace condition	tau.cr.q70.alpha-0.375★	-0.006	0.560	0.682	+0.242
779	CAST-style, trace condition	tau.cr.q70.alpha-0.75	-0.066	0.646	0.595	+0.241
780	CAST-style, trace condition	tau.cr.q50.alpha-0.5	-0.028	0.859	0.380	+0.239
781	CAST-style, trace condition	tau.cr.q50.alpha-0.75	-0.110	0.875	0.353	+0.228
782	CAST-style, trace condition	tau.cr.q90.alpha-0.5	-0.001	0.411	0.802	+0.214
783	CAST-style, trace condition	tau.cr.q90.alpha-0.75	-0.022	0.414	0.795	+0.209
784	CAST-style, trace condition	tau.cr.q90.alpha-0.375	+0.000	0.370	0.832	+0.202
785	CAST-style, trace condition	tau.cr.q30.alpha-0.25	-0.001	0.521	0.670	+0.190
786	CAST-style, trace condition	tau.cr.q50.alpha-0.25	+0.000	0.474	0.716	+0.190
787	CAST-style, trace condition	tau.cr.q30.alpha-0.5	-0.038	0.945	0.244	+0.189
788	CAST-style, trace condition	tau.cr.q70.alpha-0.25	+0.000	0.352	0.809	+0.161
789	CAST-style, trace condition	tau.cr.q30.alpha-0.75	-0.139	0.969	0.178	+0.147
790	CAST-style, trace condition	tau.cr.q90.alpha-0.25	+0.001	0.245	0.894	+0.139
791	CAST, prompt condition	L16.q50.alpha-0.375	-0.018	0.607	0.673	+0.280
792	CAST, prompt condition	L16.q30.alpha-0.375	-0.017	0.711	0.565	+0.276
793	CAST, prompt condition	L14.q30.alpha-0.375	-0.017	0.708	0.567	+0.275
794	CAST, prompt condition	L14.q50.alpha-0.375	-0.018	0.594	0.680	+0.274
795	CAST, prompt condition	L16.q50.alpha-0.5	-0.022	0.682	0.580	+0.262
796	CAST, prompt condition	L14.q50.alpha-0.5	-0.021	0.669	0.588	+0.258
797	CAST, prompt condition	L16.q50.alpha-0.75	-0.069	0.690	0.562	+0.252
798	CAST, prompt condition	L14.q50.alpha-0.75	-0.070	0.674	0.571	+0.245
799	CAST, prompt condition	L14.q30.alpha-0.5	-0.034	0.810	0.415	+0.225
800	CAST, prompt condition	L16.q30.alpha-0.5	-0.034	0.815	0.408	+0.223
801	CAST, prompt condition	L14.q30.alpha-0.75	-0.106	0.820	0.378	+0.198
802	CAST, prompt condition	L16.q30.alpha-0.75	-0.104	0.826	0.369	+0.195
803	CAST, prompt condition	L16.q30.alpha-0.25★	-0.007	0.443	0.730	+0.172
804	CAST, prompt condition	L14.q30.alpha-0.25	-0.008	0.440	0.728	+0.168
805	CAST, prompt condition	L16.q50.alpha-0.25	-0.009	0.362	0.792	+0.153
806	CAST, prompt condition	L14.q50.alpha-0.25	-0.010	0.354	0.792	+0.146
807	PTS post-hoc	q60.alpha-0.75	-0.031	0.570	0.763	+0.334
808	PTS post-hoc	q50.alpha-0.75	-0.050	0.674	0.654	+0.328
809	PTS post-hoc	q40.alpha-0.75	-0.073	0.771	0.555	+0.326
810	PTS post-hoc	q30.alpha-0.75	-0.093	0.862	0.454	+0.316
811	PTS post-hoc	q30.alpha-0.375	-0.013	0.737	0.578	+0.315
812	PTS post-hoc	q60.alpha-0.375★	-0.003	0.510	0.790	+0.300
813	PTS post-hoc	q40.alpha-0.375	-0.015	0.654	0.645	+0.299
814	PTS post-hoc	q50.alpha-0.375	-0.012	0.586	0.707	+0.293
815	PTS post-hoc	q30.alpha-0.25	+0.002	0.477	0.735	+0.212
816	PTS post-hoc	q40.alpha-0.25	+0.002	0.424	0.767	+0.191
817	PTS post-hoc	q50.alpha-0.25	+0.002	0.380	0.806	+0.186
818	PTS post-hoc	q60.alpha-0.25	+0.005	0.320	0.864	+0.184
819	PTS post-hoc	q30.ablate	+0.012	0.388	0.765	+0.153
820	PTS post-hoc	q40.ablate	+0.014	0.336	0.795	+0.131
821	PTS post-hoc	q50.ablate	+0.016	0.281	0.841	+0.122
822	PTS post-hoc	q60.ablate	+0.012	0.232	0.887	+0.119
823	PTS, prefix decision	q50.at32.alpha-0.375★	+0.004	0.573	0.737	+0.310
824	PTS, prefix decision	q50.at16.alpha-0.375	-0.002	0.573	0.731	+0.304
825	PTS, prefix decision	q30.at16.alpha-0.375	+0.002	0.729	0.572	+0.302
826	PTS, prefix decision	q50.at64.alpha-0.375	+0.017	0.562	0.739	+0.301
827	PTS, prefix decision	q30.at32.alpha-0.375	+0.002	0.690	0.602	+0.293
828	PTS, prefix decision	q50.at128.alpha-0.375	-0.001	0.547	0.708	+0.255
829	PTS, prefix decision	q30.at64.alpha-0.375	+0.010	0.656	0.595	+0.252
830	PTS, prefix decision	q30.at128.alpha-0.375	+0.000	0.651	0.574	+0.225
831	PTS, prefix decision	q50.at32	+0.019	0.245	0.894	+0.139
832	PTS, prefix decision	q30.at32	+0.018	0.320	0.816	+0.137
833	PTS, prefix decision	q50.at64	+0.002	0.214	0.898	+0.111
834	PTS, prefix decision	q30.at16	-0.002	0.326	0.781	+0.106
835	PTS, prefix decision	q50.at16	+0.003	0.224	0.878	+0.102

family	configuration	Δ acc	removal	retention	Sel
PTS, prefix decision	q50_at128	+0.000	0.182	0.912	+0.094
PTS, prefix decision	q30_at64	-0.001	0.250	0.836	+0.086
PTS, prefix decision	q30_at128	-0.003	0.214	0.871	+0.085
PTS, prompt decision	q60_alpha-0.75	-0.041	0.570	0.767	+0.337
PTS, prompt decision	q60_alpha-0.5*	-0.009	0.568	0.769	+0.336
PTS, prompt decision	q50_alpha-0.75	-0.061	0.656	0.661	+0.317
PTS, prompt decision	q50_alpha-0.5	-0.018	0.654	0.663	+0.316
PTS, prompt decision	q60_alpha-0.375	-0.015	0.518	0.790	+0.308
PTS, prompt decision	q50_alpha-0.375	-0.010	0.589	0.712	+0.301
PTS, prompt decision	q30_alpha-0.375	-0.013	0.724	0.569	+0.293
PTS, prompt decision	q40_alpha-0.5	-0.024	0.724	0.567	+0.291
PTS, prompt decision	q40_alpha-0.75	-0.084	0.729	0.562	+0.291
PTS, prompt decision	q40_alpha-0.375	-0.012	0.646	0.643	+0.289
PTS, prompt decision	q30_alpha-0.5	-0.020	0.826	0.463	+0.288
PTS, prompt decision	q30_alpha-0.75	-0.099	0.839	0.445	+0.284
PTS, prompt decision	q30_alpha-0.25	+0.007	0.464	0.735	+0.199
PTS, prompt decision	q60_alpha-0.25	-0.002	0.320	0.871	+0.191
PTS, prompt decision	q40_alpha-0.25	+0.004	0.409	0.781	+0.190
PTS, prompt decision	q50_alpha-0.25	+0.002	0.365	0.825	+0.190
earlier gate + ablation	u-crq50-ablate*	+0.007	0.378	0.740	+0.118
ungated actions	ablate	+0.007	0.451	0.691	+0.141
ungated actions	clamp.q50	+0.017	0.307	0.814	+0.122
ungated actions	clamp.q30	+0.012	0.339	0.763	+0.102
ungated actions	otg-cal	+0.004	0.115	0.926	+0.040

Table 6: Tuning split, Qwen2.5-7B-Instruct ($n = 1,200$, M5).

family	configuration	Δ acc	removal	retention	Sel
additive CAA	alpha-0.125*	-0.026	0.212	0.911	+0.123
additive CAA	alpha-0.375	-0.050	0.208	0.892	+0.100
additive CAA	alpha-0.25	-0.044	0.200	0.887	+0.087
additive CAA	alpha-0.5	-0.076	0.204	0.855	+0.059
additive CAA	alpha-0.75	-0.138	0.233	0.759	-0.007
additive CAA	alpha-1.0	-0.217	0.412	0.544	-0.043
additive CAA	alpha-1.25	-0.334	0.567	0.272	-0.161
additive CAA	alpha-1.5	-0.473	0.375	0.230	-0.395
MiMiC	L18.reg0.0001	-0.013	0.279	0.906	+0.185
MiMiC	L18.reg0.01*	-0.007	0.263	0.911	+0.173
MiMiC	L14.reg0.01	-0.018	0.254	0.912	+0.166
MiMiC	L14.reg0.0001	-0.023	0.233	0.912	+0.145
MiMiC	L10.reg0.1	-0.019	0.250	0.890	+0.140
MiMiC	L14.reg0.1	-0.021	0.217	0.923	+0.140
MiMiC	L18.reg0.1	-0.013	0.233	0.906	+0.139
MiMiC	L10.reg0.01	-0.022	0.237	0.892	+0.129
MiMiC	L10.reg0.0001	-0.031	0.233	0.868	+0.101
Linear-AcT	L18.lam1.0*	-0.004	0.192	0.948	+0.140
Linear-AcT	L14.lam1.0	+0.002	0.175	0.955	+0.130
Linear-AcT	L14.lam0.25	+0.004	0.138	0.969	+0.106
Linear-AcT	L10.lam1.0	-0.008	0.158	0.947	+0.105
Linear-AcT	L18.lam0.25	-0.006	0.138	0.957	+0.095
Linear-AcT	L10.lam0.25	-0.013	0.133	0.958	+0.092
Linear-AcT	L18.lam0.5	-0.006	0.142	0.950	+0.091
Linear-AcT	L10.lam0.5	-0.013	0.125	0.951	+0.076
Linear-AcT	L14.lam0.5	-0.005	0.113	0.961	+0.073
CAST-style, trace condition	tau.cr.q30.alpha-0.375	-0.043	0.188	0.907	+0.094
CAST-style, trace condition	tau.cr.q30.alpha-0.25	-0.035	0.183	0.906	+0.089
CAST-style, trace condition	tau.cr.q50.alpha-0.375	-0.040	0.158	0.918	+0.076
CAST-style, trace condition	tau.cr.q50.alpha-0.25	-0.033	0.154	0.919	+0.074
CAST-style, trace condition	tau.cr.q30.alpha-0.5	-0.067	0.188	0.879	+0.067
CAST-style, trace condition	tau.cr.q70.alpha-0.375	-0.024	0.108	0.945	+0.053
CAST-style, trace condition	tau.cr.q50.alpha-0.5	-0.059	0.154	0.898	+0.052
CAST-style, trace condition	tau.cr.q70.alpha-0.25	-0.022	0.104	0.945	+0.049
CAST-style, trace condition	tau.cr.q50.alpha-0.75	-0.079	0.188	0.858	+0.045
CAST-style, trace condition	tau.cr.q90.alpha-0.25	-0.010	0.067	0.975	+0.041
CAST-style, trace condition	tau.cr.q70.alpha-0.5	-0.036	0.104	0.937	+0.041
CAST-style, trace condition	tau.cr.q90.alpha-0.375*	-0.009	0.062	0.979	+0.041
CAST-style, trace condition	tau.cr.q90.alpha-0.5	-0.018	0.071	0.970	+0.041
CAST-style, trace condition	tau.cr.q70.alpha-0.75	-0.048	0.125	0.912	+0.037
CAST-style, trace condition	tau.cr.q30.alpha-0.75	-0.098	0.212	0.824	+0.036
CAST-style, trace condition	tau.cr.q90.alpha-0.75	-0.023	0.079	0.956	+0.035
CAST, prompt condition	L14.q30.alpha-0.25*	-0.032	0.167	0.916	+0.082
CAST, prompt condition	L16.q30.alpha-0.25	-0.032	0.167	0.916	+0.082
CAST, prompt condition	L14.q30.alpha-0.375	-0.040	0.163	0.918	+0.081
CAST, prompt condition	L16.q30.alpha-0.375	-0.040	0.163	0.918	+0.081
CAST, prompt condition	L16.q50.alpha-0.375	-0.034	0.142	0.931	+0.072
CAST, prompt condition	L14.q50.alpha-0.375	-0.034	0.142	0.929	+0.071
CAST, prompt condition	L14.q50.alpha-0.25	-0.026	0.138	0.932	+0.069
CAST, prompt condition	L16.q50.alpha-0.25	-0.027	0.133	0.932	+0.065

family	configuration	Δ acc	removal	retention	Sel
CAST, prompt condition	L14.q30.alpha-0.5	-0.058	0.167	0.890	+0.057
CAST, prompt condition	L16.q30.alpha-0.5	-0.058	0.167	0.890	+0.057
CAST, prompt condition	L16.q50.alpha-0.5	-0.050	0.133	0.914	+0.048
CAST, prompt condition	L14.q50.alpha-0.5	-0.051	0.133	0.913	+0.046
CAST, prompt condition	L16.q50.alpha-0.75	-0.078	0.158	0.868	+0.026
CAST, prompt condition	L14.q50.alpha-0.75	-0.079	0.150	0.868	+0.018
CAST, prompt condition	L14.q30.alpha-0.75	-0.105	0.188	0.821	+0.009
CAST, prompt condition	L16.q30.alpha-0.75	-0.105	0.183	0.822	+0.006
PTS post-hoc	q40.ablate*	-0.006	0.254	0.927	+0.181
PTS post-hoc	q50.ablate	-0.002	0.233	0.940	+0.173
PTS post-hoc	q30.ablate	-0.009	0.254	0.918	+0.172
PTS post-hoc	q60.ablate	-0.003	0.212	0.948	+0.161
PTS post-hoc	q40.alpha-0.375	-0.044	0.183	0.907	+0.090
PTS post-hoc	q30.alpha-0.375	-0.048	0.192	0.898	+0.090
PTS post-hoc	q40.alpha-0.25	-0.043	0.183	0.902	+0.085
PTS post-hoc	q60.alpha-0.375	-0.032	0.154	0.929	+0.084
PTS post-hoc	q30.alpha-0.25	-0.045	0.192	0.892	+0.083
PTS post-hoc	q60.alpha-0.25	-0.028	0.150	0.931	+0.081
PTS post-hoc	q50.alpha-0.375	-0.039	0.163	0.916	+0.078
PTS post-hoc	q50.alpha-0.25	-0.037	0.154	0.912	+0.066
PTS post-hoc	q60.alpha-0.75	-0.062	0.158	0.883	+0.041
PTS post-hoc	q50.alpha-0.75	-0.083	0.179	0.854	+0.033
PTS post-hoc	q40.alpha-0.75	-0.098	0.204	0.826	+0.030
PTS post-hoc	q30.alpha-0.75	-0.110	0.217	0.805	+0.021
PTS, prefix decision	q30.at32	-0.010	0.163	0.951	+0.113
PTS, prefix decision	q30.at32.alpha-0.375	-0.012	0.183	0.929	+0.113
PTS, prefix decision	q30.at16*	-0.008	0.163	0.948	+0.111
PTS, prefix decision	q50.at32	-0.007	0.142	0.966	+0.102
PTS, prefix decision	q50.at16	-0.006	0.138	0.960	+0.097
PTS, prefix decision	q30.at64	-0.008	0.133	0.951	+0.084
PTS, prefix decision	q30.at16.alpha-0.375	-0.030	0.167	0.916	+0.082
PTS, prefix decision	q50.at32.alpha-0.375	-0.013	0.133	0.943	+0.077
PTS, prefix decision	q30.at64.alpha-0.375	-0.014	0.133	0.942	+0.075
PTS, prefix decision	q50.at64	-0.005	0.108	0.963	+0.072
PTS, prefix decision	q30.at128.alpha-0.375	-0.005	0.083	0.969	+0.052
PTS, prefix decision	q50.at128.alpha-0.375	-0.004	0.079	0.972	+0.051
PTS, prefix decision	q50.at16.alpha-0.375	-0.029	0.113	0.938	+0.051
PTS, prefix decision	q50.at64.alpha-0.375	-0.018	0.083	0.955	+0.038
PTS, prefix decision	q30.at128	+0.002	0.058	0.979	+0.037
PTS, prefix decision	q50.at128	+0.002	0.050	0.977	+0.027
PTS, prompt decision	q30.alpha-0.375*	-0.045	0.204	0.901	+0.105
PTS, prompt decision	q30.alpha-0.25	-0.042	0.196	0.899	+0.095
PTS, prompt decision	q40.alpha-0.375	-0.043	0.188	0.907	+0.094
PTS, prompt decision	q50.alpha-0.375	-0.040	0.171	0.916	+0.086
PTS, prompt decision	q40.alpha-0.25	-0.043	0.175	0.906	+0.081
PTS, prompt decision	q50.alpha-0.25	-0.038	0.163	0.914	+0.077
PTS, prompt decision	q60.alpha-0.375	-0.033	0.138	0.935	+0.072
PTS, prompt decision	q30.alpha-0.5	-0.068	0.204	0.868	+0.072
PTS, prompt decision	q60.alpha-0.25	-0.031	0.133	0.929	+0.063
PTS, prompt decision	q50.alpha-0.5	-0.056	0.163	0.897	+0.059
PTS, prompt decision	q40.alpha-0.5	-0.068	0.175	0.874	+0.049
PTS, prompt decision	q60.alpha-0.5	-0.047	0.129	0.914	+0.044
PTS, prompt decision	q50.alpha-0.75	-0.080	0.179	0.855	+0.034
PTS, prompt decision	q60.alpha-0.75	-0.067	0.146	0.884	+0.030
PTS, prompt decision	q30.alpha-0.75	-0.108	0.217	0.810	+0.026
PTS, prompt decision	q40.alpha-0.75	-0.101	0.196	0.826	+0.022
earlier gate + ablation	u.crq50.ablate*	-0.008	0.196	0.938	+0.134
ungated actions	ablate	-0.008	0.258	0.913	+0.171
ungated actions	clamp.q30	-0.015	0.221	0.918	+0.139
ungated actions	otq.cal	-0.003	0.154	0.962	+0.116
ungated actions	clamp.q50	-0.008	0.183	0.928	+0.112

D THE SELECTIVITY IDENTITY IN PRACTICE

Proposition 1 needs the output to be unchanged wherever the decision does not fire. For the post-hoc and prompt decisions this holds by construction, so a gated arm equals, question by question, either its unsteered row or its ungated row. Composing the two requires one ungated run per action and one set of decision scores, and it differs from a real gated run only through batched-decoding nondeterminism: a flagged question is generated in a different batch in the two cases. Table 7 compares the composition with the real run for the four gated arms of each confirmatory split, and reports the prediction from the ungated conversion rates (independence within each state).

Across the twelve arms the composition differs from the real run by at most 0.025 (Qwen, PTS post-hoc) and by 0.005 at the median. The prediction is higher than the real run in eleven of

setting	arm	TPR	FPR	ρ_O	ρ_C	predicted	composed	real run
Gemma, MMLU	PTS post-hoc	0.541	0.222	0.841	0.585	+0.325	+0.296	+0.294
	PTS at the prompt	0.549	0.242	0.981	0.872	+0.328	+0.306	+0.306
	CAST-style, trace	0.603	0.380	0.841	0.585	+0.285	+0.244	+0.248
	CAST, prompt	0.805	0.635	0.559	0.339	+0.234	+0.207	+0.217
Qwen, MMLU	PTS post-hoc	0.869	0.478	0.240	0.084	+0.168	+0.148	+0.172
	PTS at the prompt	0.920	0.594	0.247	0.111	+0.161	+0.126	+0.120
	CAST-style, trace	0.286	0.101	0.247	0.111	+0.059	+0.030	+0.040
	CAST, prompt	0.817	0.670	0.277	0.098	+0.161	+0.143	+0.122
Gemma, ARC	PTS post-hoc	0.675	0.405	0.513	0.219	+0.258	+0.239	+0.235
	PTS at the prompt	0.417	0.247	0.991	0.896	+0.191	+0.181	+0.182
	CAST-style, trace	0.912	0.767	0.513	0.219	+0.300	+0.297	+0.289
	CAST, prompt	0.386	0.379	0.513	0.219	+0.115	+0.107	+0.109

Table 7: Real gated runs against the composition of ungated runs and against the prediction TPR ρ_O – FPR ρ_C with ungated conversion rates.

twelve rows, and the same bias appears on the tuning split (Fig. 2a). Flagged answers are converted more often than the average answer of their state in both states: for PTS post-hoc, 0.91 of flagged overconfident-wrong answers change state against 0.84 ungated, and 0.89 of flagged confident-right answers against 0.59 ungated. The larger increase on the protected state lowers selectivity below the independence prediction. The composition, which uses no such assumption, is what scores the gated configurations during selection.

E SECONDARY READOUTS

Table 8 reports the Gemma confirmatory arms under the two secondary readouts. The letter-logit confidence M2 assigns confidence above 0.5 to nearly every answer, so its OCW stratum is close to the set of wrong answers and every method moves it little. Under M4 the edits act on four separate verification generations; PTS is level with the ungated edits there, which is expected for a decision fitted to M5 labels.

method (Gemma, confirmatory)	M2 (letter-logit)		M4 (verification)	
	Sel [95% CI]	Δ vs PTS	Sel [95% CI]	Δ vs PTS
prompting	+0.083 [+0.052, +0.113]	−0.008 [−0.045, +0.028]	+0.138 [+0.086, +0.191]	−0.102 [−0.167, −0.034]
additive CAA	+0.089 [+0.059, +0.120]	−0.001 [−0.036, +0.034]	+0.242 [+0.187, +0.296]	+0.003 [−0.061, +0.065]
dir. ablation	+0.092 [+0.062, +0.122]	+0.002 [−0.034, +0.036]	+0.246 [+0.193, +0.299]	+0.007 [−0.054, +0.070]
MiMiC	+0.064 [+0.036, +0.092]	−0.027 [−0.063, +0.008]	+0.244 [+0.189, +0.298]	+0.004 [−0.059, +0.067]
Linear-AcT	+0.029 [+0.009, +0.050]	−0.061 [−0.091, −0.031]	+0.111 [+0.068, +0.155]	−0.128 [−0.185, −0.070]
CAST (prompt)	+0.073 [+0.045, +0.102]	−0.017 [−0.052, +0.017]	+0.207 [+0.155, +0.259]	−0.032 [−0.092, +0.027]
CAST-style (trace)	+0.074 [+0.046, +0.103]	−0.016 [−0.041, +0.010]	+0.224 [+0.173, +0.277]	−0.015 [−0.063, +0.033]
PTS, post-hoc	+0.090 [+0.064, +0.118]	—	+0.239 [+0.192, +0.286]	—
PTS, prompt	+0.099 [+0.069, +0.128]	+0.008 [−0.023, +0.038]	+0.245 [+0.198, +0.292]	+0.006 [−0.039, +0.052]

Table 8: Secondary readouts on the confirmatory split; paired bootstrap intervals, no multiplicity adjustment.

method (Qwen, confirmatory)	M2 (letter-logit)		M4 (verification)	
	Sel [95% CI]	Δ vs PTS	Sel [95% CI]	Δ vs PTS
prompting	+0.132 [+0.102, +0.163]	+0.013 [−0.021, +0.046]	+0.178 [+0.144, +0.213]	+0.073 [+0.033, +0.114]
additive CAA	+0.118 [+0.089, +0.149]	−0.000 [−0.033, +0.034]	+0.096 [+0.060, +0.132]	−0.009 [−0.050, +0.033]
dir. ablation	+0.119 [+0.089, +0.150]	+0.000 [−0.024, +0.025]	+0.192 [+0.159, +0.225]	+0.087 [+0.057, +0.117]
MiMiC	+0.130 [+0.100, +0.160]	+0.011 [−0.023, +0.045]	+0.104 [+0.073, +0.136]	−0.001 [−0.040, +0.038]
Linear-AcT	+0.099 [+0.072, +0.127]	−0.020 [−0.053, +0.013]	+0.078 [+0.051, +0.106]	−0.026 [−0.062, +0.009]
CAST (prompt)	+0.110 [+0.081, +0.140]	−0.009 [−0.043, +0.025]	+0.159 [+0.125, +0.193]	+0.054 [+0.012, +0.096]
CAST-style (trace)	+0.030 [+0.012, +0.048]	−0.089 [−0.120, −0.059]	+0.121 [+0.096, +0.145]	+0.016 [−0.021, +0.053]
PTS, post-hoc	+0.119 [+0.090, +0.149]	—	+0.105 [+0.076, +0.135]	—
PTS, prompt	+0.096 [+0.063, +0.128]	−0.023 [−0.059, +0.012]	+0.170 [+0.134, +0.206]	+0.065 [+0.022, +0.108]

Table 9: Secondary readouts on the Qwen confirmatory split.

Table 10 gives the full state transitions of the frozen PTS arms under M5. Selectivity counts only the first and last rows; the table also shows the answers the edit corrects or breaks and the few answers it turns overconfident.

Table 11 reports calibration of the stated confidence on the confirmatory split: expected calibration error (10 bins), Brier score, and the AUROC with which the confidence separates right from wrong answers.

Where the edit acts. Rerunning the frozen PTS decision with the action applied only during the reasoning trace or only during the confidence turn gives, on Gemma (shift), 0.159 and 0.274

	before \ after	OCW	NCW	NCR	CR
Gemma, PTS	OCW	489	350	118	6
	NCW	10	298	51	2
	NCR	2	61	207	7
	CR	11	95	171	1122
Qwen, PTS	OCW	425	43	10	85
	NCW	54	121	23	27
	NCR	21	25	115	44
	CR	81	25	40	1861

Table 10: State transitions of the frozen PTS arms on the confirmatory split (counts of questions).

arm	Gemma-2-2B			Qwen2.5-7B		
	ECE ↓	Brier ↓	AUROC ↑	ECE ↓	Brier ↓	AUROC ↑
unsteered	0.354	0.372	0.643	0.215	0.230	0.741
shift −0.25, ungated	0.202	0.281	0.659	0.238	0.245	0.741
ablation, ungated	0.243	0.296	0.653	0.220	0.232	0.743
PTS, post-hoc	0.255	0.294	0.689	0.215	0.229	0.740
override, probe on answer	0.234	0.285	0.684	0.139	0.195	0.760
override, CAST condition	0.197	0.297	0.646	0.194	0.222	0.714

Table 11: Calibration of the stated confidence (M5) on the confirmatory split.

[0.237, 0.312] against 0.294 for both turns, and on Qwen (ablation) 0.172 and 0.006 against 0.172 (exploratory).

F NET EFFECTS OF EVERY ARM

Selectivity is computed on the answers that start in OCW or CR; it does not see answers that enter OCW from another state. Table 12 reports, for every confirmatory arm and the three null arms of Appendix A, selectivity net of the no-edit null, the change in the share of OCW answers among all questions, the change in the rate of confident answers among wrong and among right answers, Δ ECE, Δ Brier, the change in the share of traces without a `\boxed{ }` answer (which then receive a forced box, Appendix O), and selectivity recomputed without the questions whose steered trace has no box. The ungated rerun reproduces the unsteered outputs exactly, because it uses the same batches; the gated no-edit null regenerates the flagged questions in different batches, and the resulting nondeterminism alone yields selectivity 0.049 (Gemma), 0.091 (Qwen) and 0.016 (ARC). Excluding traces without a box leaves the ordering on Gemma unchanged: PTS post-hoc 0.257, CAST-style 0.233, additive 0.215.

G REFERENCE ARMS FOR EVERY CONDITIONAL METHOD

Table 13 applies the three references of Sec. 3 to the four conditional multiple-choice arms of each setting, with ten random directions per arm (protocol v22). The matched override lowers the stated confidence of exactly the answers the arm’s own decision flags. PTS post-hoc passes all six of its registered null tests, and 18 of the 24 tests pass after Holm adjustment. Eq. (3), evaluated with ungated conversion rates before the gated runs, predicts the sign of steer minus override in all rows where the predicted difference exceeds 0.021. The CAST conditions fire on many protected answers and their shift spares most of them, so steering is more selective than overriding with them. Table 14 reports the hundred-direction test of protocol v23 for PTS post-hoc.

H FREE-TEXT BEHAVIOR: TOXIC CONTINUATION

Data and readout. Prompts are RealToxicityPrompts prompts whose own toxicity is at least 0.5 (21,744), shuffled with seed 0 and split into extraction (1,000), probe pool (3,000), tuning (1,000) and confirmation (3,000). Each prompt is rendered with the chat template as “Continue the following text.” and continued for 40 greedy tokens. The judge is `s-nlp/roberta-toxicity-classifier`; a continuation is toxic at probability 0.5 or above, refused when it contains a refusal phrase in its first 200 characters or is degenerate (fewer than three

arm	Sel	Sel-null	Δ OCW (pp)	$\Delta P(\text{conf} \mid \text{wrong})$ (pp)	$\Delta P(\text{conf} \mid \text{right})$ (pp)	Δ ECE	Δ Brier	Δ no box (pp)	Sel, boxed only
<i>Gemma-2-2B, MMLU</i>									
prompting	0.200	+0.151	-5.9	-12.7	-6.1	-0.030	-0.032	+1.1	0.185
additive CAA	0.220	+0.171	-13.0	-29.4	-18.9	-0.153	-0.091	+1.3	0.215
dir. ablation	0.173	+0.124	-8.7	-18.6	-13.0	-0.111	-0.076	+0.4	0.175
MiMiC	0.117	+0.068	-2.1	-4.7	-4.6	-0.018	-0.020	+0.0	0.106
Linear-AcT	0.061	+0.012	-0.2	+0.4	+0.6	-0.003	-0.006	-0.1	0.053
CAST (prompt)	0.217	+0.168	-11.2	-25.6	-13.4	-0.137	-0.086	+0.6	0.206
CAST-style (trace)	0.248	+0.199	-16.6	-38.0	-23.1	-0.094	-0.067	+8.2	0.233
PTS, post-hoc	0.294	+0.245	-15.0	-33.8	-16.0	-0.099	-0.079	+5.7	0.257
PTS, prefix	0.294	+0.245	-17.2	-39.5	-20.2	-0.136	-0.091	+2.0	0.284
PTS, prompt	0.306	+0.257	-17.6	-40.8	-18.7	-0.084	-0.077	+15.0	0.247
null: decision, no edit	0.049	+0.000	+0.0	+0.5	+0.9	-0.005	-0.007	-0.4	0.037
null: decision, random dir.	0.215	+0.166	-8.9	-19.6	-7.9	-0.077	-0.061	+7.2	0.185
rerun, no edit	0.000	-0.049	+0.0	+0.0	+0.0	+0.000	+0.000	+0.0	0.000
<i>Qwen2.5-7B, MMLU</i>									
prompting	0.212	+0.121	-1.3	-5.6	-2.0	-0.004	-0.002	+3.6	0.196
additive CAA	0.151	+0.060	+2.9	+6.5	+3.5	+0.012	+0.008	-0.8	0.149
dir. ablation	0.156	+0.065	+1.3	+2.3	+0.8	+0.005	+0.001	+0.6	0.156
MiMiC	0.154	+0.063	+1.4	+4.9	+0.2	+0.020	+0.014	+0.5	0.151
Linear-AcT	0.123	+0.032	+0.6	+1.4	+0.1	+0.009	+0.004	+0.7	0.114
CAST (prompt)	0.122	+0.031	+3.7	+10.4	+3.0	+0.027	+0.021	+0.5	0.124
CAST-style (trace)	0.040	-0.051	+2.4	+7.2	+2.4	+0.013	+0.009	+0.2	0.036
PTS, post-hoc	0.172	+0.081	+0.6	+1.6	+0.7	-0.000	-0.002	+0.5	0.166
PTS, prefix	0.109	+0.018	+1.0	+1.6	-0.6	+0.013	+0.010	+0.1	0.092
PTS, prompt	0.120	+0.029	+6.9	+15.2	+5.4	+0.049	+0.039	+1.1	0.123
null: decision, no edit	0.091	+0.000	-0.1	-0.9	-0.2	+0.000	-0.000	+0.1	0.084
null: decision, random dir.	0.119	+0.028	-0.1	-1.6	+0.4	-0.002	-0.004	-0.2	0.117
rerun, no edit	0.000	-0.091	+0.0	+0.0	+0.0	+0.000	+0.000	+0.0	0.000
<i>Gemma-2-2B, ARC</i>									
prompting	0.238	+0.222	-1.8	-6.1	-3.7	+0.009	+0.005	-0.3	0.234
additive CAA	0.294	+0.278	-5.9	-23.3	-11.4	-0.069	-0.013	+1.2	0.289
dir. ablation	0.215	+0.198	-3.5	-15.6	-8.6	-0.055	-0.007	+1.0	0.216
MiMiC	0.198	+0.182	-1.7	-3.8	-3.6	-0.007	-0.003	+0.7	0.198
Linear-AcT	0.109	+0.093	-0.1	+0.4	+0.7	+0.000	-0.001	+0.3	0.108
CAST (prompt)	0.109	+0.093	-2.0	-8.6	-5.3	-0.028	-0.002	+0.3	0.090
CAST-style (trace)	0.289	+0.273	-5.4	-20.9	-10.7	-0.061	-0.015	+1.6	0.276
PTS, post-hoc	0.235	+0.219	-3.8	-16.2	-5.4	-0.060	-0.025	+1.0	0.226
PTS, prefix	0.276	+0.260	-9.2	-37.1	-17.6	+0.000	+0.003	+1.5	0.267
PTS, prompt	0.182	+0.166	-8.1	-35.8	-18.4	+0.028	+0.019	+10.6	0.150
null: decision, no edit	0.016	+0.000	+1.7	+5.3	+0.5	+0.011	+0.011	+0.3	0.020
null: decision, random dir.	0.194	+0.178	-3.7	-15.0	-6.6	-0.006	-0.001	+2.7	0.176
rerun, no edit	0.000	-0.016	+0.0	+0.0	+0.0	+0.000	+0.000	+0.0	0.000

Table 12: Net effects on the confirmatory splits (point estimates; intervals for the main-text quantities are given there). “null” is the frozen decision with no edit.

setting	method	Sel	no-edit null		random-edit null (10)		matched override		
			Sel	Δ	mean Sel	Δ	Sel	steer	override [95% CI]
Gemma, MMLU	PTS, post-hoc	0.294	0.049	+0.245*	0.141	+0.153*	0.319	-0.025	[-0.04, -0.01]
	PTS, prompt	0.306	0.041	+0.265*	0.147	+0.159*	0.307	-0.001	[-0.00, +0.00]
	CAST-style (trace)	0.248	0.049	+0.199*	0.112	+0.136*	0.223	+0.025	[+0.00, +0.05]
	CAST (prompt)	0.217	0.044	+0.173*	0.105	+0.112*	0.169	+0.048	[+0.01, +0.09]
Qwen, MMLU	PTS, post-hoc	0.172	0.091	+0.081*	0.122	+0.051*	0.391	-0.218	[-0.26, -0.17]
	PTS, prompt	0.120	0.077	+0.043	0.224	-0.104	0.326	-0.206	[-0.25, -0.16]
	CAST-style (trace)	0.040	0.035	+0.005	0.080	-0.040	0.185	-0.145	[-0.18, -0.11]
	CAST (prompt)	0.122	0.065	+0.057*	0.182	-0.059	0.147	-0.025	[-0.07, +0.02]
Gemma, ARC	PTS, post-hoc	0.235	0.016	+0.219*	0.165	+0.070*	0.270	-0.035	[-0.10, +0.03]
	PTS, prompt	0.182	0.045	+0.137*	0.119	+0.063*	0.169	+0.013	[+0.01, +0.02]
	CAST-style (trace)	0.289	0.077	+0.213*	0.211	+0.078*	0.146	+0.144	[+0.07, +0.22]
	CAST (prompt)	0.109	0.020	+0.089*	0.078	+0.032	0.006	+0.103	[+0.04, +0.16]

Table 13: Reference arms on the confirmatory splits. Δ is the method minus the reference; *: Holm-adjusted $p < 0.05$ over the 24 registered tests.

words or a distinct bigram ratio below one half), and clean otherwise. Unsteered, Gemma produces 451 toxic and 1,910 clean continuations on the confirmation prompts and Qwen 608 and 1,907. Selectivity is the share of toxic continuations that are no longer toxic minus the share of clean continuations lost, where a clean continuation is kept when the steered one is clean and is identical to, or has `all-mpnet-base-v2` cosine at least 0.5 with, the unsteered one. On the Gemma tuning split, this criterion keeps 84% of clean continuations at the smallest shift and 36% at the largest, so rewriting a continuation into different content is counted as damage.

Operator. The direction is the difference of mean layer-14 activations over toxic and clean continuations of the extraction split, and the dose is a multiple of the mean activation norm. The prompt decision is a logistic probe on the mean prompt activation (layer 18 for Gemma, 10 for Qwen; tuning AUROC 0.808 and 0.823), the post-hoc decision a probe on the unsteered continuation (0.946 and 0.890), and the CAST condition the cosine of the mean prompt activation to the toxic-minus-clean difference of the probe pool. The decision gates each row inside the original batch, so an unflagged row is generated exactly as in the unsteered run. The tuning grid holds five shift doses, ablation and prompting, and three thresholds for each decision; the selection rule is the highest tuning selectivity.

setting	PTS	random mean	95th pct.	max	directions $\geq$ PTS	PTS - mean [95% CI]
Gemma, MMLU (100)	0.294	0.131	0.237	0.293	0	+0.163 [+0.140, +0.188]
Gemma, ARC (100)	0.235	0.167	0.230	0.257	2	+0.068 [+0.017, +0.119]
Qwen, MMLU (30, matched)	0.172	0.158	0.175	0.178	3	+0.014 [-0.017, +0.045]

Table 14: PTS post-hoc against random unit directions sent through its own decision at its own dose (Gemma and ARC, 100 directions) and, for the ablation selected on Qwen, against 30 random directions displaced as far as the ablation displaces the confidence direction ($h - \langle h, \hat{v} \rangle r$).

Results. Table 4 gives the confirmatory comparison. Nine of the ten registered tests pass after Holm adjustment. Refusing on a decision converts every flagged clean continuation, so by Eq. (3) a gated edit beats it whenever it spares enough of them; Eq. (3) with conversion rates from the tuning split predicts the sign of steer minus refuse for all six gated arms. Against a hundred random directions through the same decision, PTS lies above the 95th percentile (0.311 on Gemma, 0.479 on Qwen).

Changes after the v23 freeze. Two edits to the analysis code followed the freeze and precede none of the registered quantities: the conversion rates that feed the Eq. (3) prediction are taken from the tuning split, because the confirmatory split has no ungated run of the post-hoc and CAST actions, and a loading routine was rewritten for speed without changing any number. Data, judge, retention criterion, arms and the ten hypotheses are as frozen.

Quantile transport map. The map sends the per-token projection on the toxicity direction from its law over toxic continuations to its law over clean continuations of the extraction split (41 quantiles), and PTS applies $h + \lambda V^\top (S(Vh) - Vh)$ on the rows its prompt decision flags. On the Gemma tuning split $\lambda = 1, 2, 3, 4, 6$ gives selectivity 0.373, 0.577, 0.605, 0.586 and 0.552, against 0.594 for the tuned shift, so the selection rule picks $\lambda = 3$. For $\lambda > 1$ the edit extrapolates the displacement of the map and is not itself monotone on the whole source law, so it is an edit along the transport displacement rather than the map of Theorem 1, which ($\lambda = 1$) reaches 0.373. Its mean displacement on toxic tokens equals a shift of $-0.432\|h\|$. On the held-out prompts it reaches 0.584 [0.542, 0.624] against 0.540 [0.495, 0.584] for that matched shift, removes 79% of toxic continuations and keeps 79% of clean ones. On Qwen the rule keeps the shift.

Third family. Mistral-7B-Instruct-v0.3 follows the same pipeline with its own direction (layer 14), probes (prompt probe at layer 14, tuning AUROC 0.787) and tuning. Unsteered, it produces 102 toxic and 2,844 clean continuations on the confirmation prompts. PTS with the prompt decision reaches 0.520 [0.430, 0.605], removes 74% of toxic continuations and keeps 78% of clean ones; refusal on the same decision reaches 0.305, and a hundred random directions average 0.247 with a 95th percentile of 0.343. Post-hoc PTS reaches 0.712 and lowers the toxic rate to 0.4%.

Edit on generated positions only. In the prompt-decision arms the decision reads the prompt in a forward pass of its own and the edit then acts on every position. A variant that starts the edit at the first generated token, so that decision and edit share one pass, reaches 0.269 [0.223, 0.315] on Gemma and 0.460 [0.418, 0.502] on Qwen at the dose selected for the full edit, above prompting and twenty random directions in both families (exploratory, not registered).

I FRESH FREE-TEXT SPLIT

Protocol v26 (pushed before any steered rollout on the split) evaluates, on prompts 8000 to 11000 of the shuffled RealToxicityPrompts pool, the member of the PTS action family selected on the tuning split (prompt probe with a token-gated shift on Gemma and Qwen, with a shift on Mistral) against a DSAS-style token-gated shift without the input decision, a single-layer Linear-AcT per-neuron affine transport, the global shift, prompting, CAST and refusal on the PTS decision. The monotone quantile transport map (target law of clean continuations shifted by $-0.375\|h\|$) is compared on Gemma with a gated shift of equal mean displacement. Table 15 gives the selectivities; the no-edit null leaves all continuations unchanged in each family. Against refusal and against the random directions, PTS passes in all three families; against prompting it passes in all three; against the global shift it passes on Qwen; against the token gate it does not pass in any family (Gemma -0.048 , Qwen $+0.008$, Mistral $+0.087$, $p = 0.08$); the monotone map and the equal-displacement shift are level on Gemma ($+0.003$).

arm (fresh split)	Gemma-2-2B	Qwen2.5-7B	Mistral-7B
PTS (selected member)	0.549	0.602	0.496
PTS, shift (v23 arm)	0.537	0.527	0.496
token-gated shift (DSAS-style)	0.597	0.594	0.410
Linear-AcT	0.431	0.205	0.260
global shift	0.521	0.402	0.506
prompting	0.087	0.440	0.326
CAST	0.521	0.439	0.489
refusal on the PTS decision	0.397	0.454	0.322
random direction (mean of 100)	0.118	0.172	0.264
monotone transport map / equal-displacement shift	0.538 / 0.535	—	—

Table 15: Selectivity on the fresh free-text split (3,000 prompts per model; 429, 604 and 88 toxic unsteered continuations).

J PTS IN THE SETTING OF PCHI

Setting. We follow Li et al. (2026): Qwen3-4B-Instruct-2507, the JSON self-evaluation prompt of their Appendix A.1 verbatim, and greedy decoding with at most 768 new tokens. Problems are the first 10,000 unique problems of the OpenMathInstruct-2 train_1M split, the first 5,000 for fitting and the rest held out; responses whose JSON does not parse are dropped, leaving 3,137 and 3,183. An answer is correct when it matches the reference after normalization, and the confidence readout is the yes-no logit gap at the `is_confident` value. On the held-out answers this gives 1,178 wrong-confident (WY) and 1,796 correct-confident (CY) readouts; PCHI’s own sample is larger and more accurate, so absolute rates differ from theirs. The decision is a calibrated diagonal-LDA probe on the layer-18 readout state (fitted on 70% of the fitting answers, calibrated on 30%; held-out AUROC 0.81), firing at probability 0.5. PCHI learns one coefficient for each of the 128 heads of layers 19 to 22 with its published objective and hyper-parameters (200 steps, batch 8, learning rate 0.04, L1 weight 0.05, ratio 0.7) and margin -1 . PTS shifts the layer-18 readout state along the difference of mean states of wrong answers stated without and with confidence. Doses were chosen on the fitting answers (Appendix A, v24).

arm (held-out, $n_{WY}=1,178$, $n_{CY}=1,796$)	WY corrected	CY damaged	Sel [95% CI]	ECE
unsteered	—	—	0.000	0.372
shift, every answer (dose 2)	0.099	0.007	0.092 [0.075, 0.110]	0.337
PCHI, learned heads	0.399	0.045	0.354 [0.325, 0.384]	0.234
PCHI, permuted coefficients (mean of 20)	—	—	0.033	0.363
PTS (probe-gated shift, dose 8)	0.582	0.143	0.438 [0.406, 0.471]	0.188
same probe, random direction (mean of 100)	—	—	0.317	0.254
same probe, override of the readout	0.582	0.143	0.438	0.238

Table 16: The setting of PCHI on held-out answers. All three registered tests pass: PTS above PCHI (+0.084 [0.058, 0.110]), PTS above the global shift, and PCHI above its permuted coefficients.

At dose 8 the PTS shift turns every flagged readout to *no*; unlike an override of the readout it moves the stated probability rather than replacing it, and it reaches the lowest calibration error of all arms (0.188). At this dose an edit of the readout state along a random direction also flips many flagged readouts: 45 of 100 random directions reach the same selectivity, with a mean calibration error of 0.254. The learned coefficients of PCHI carry its effect: permuting them across heads lowers its selectivity to 0.033.

K LATENCY

Toxicity setting, Gemma, 750 confirmation prompts, 40 new tokens, batch 128, one idle H100, bfloat16: unsteered generation 5.79 s; generation with the gated edit 5.84 s; the forward pass over the prompts that reads the decision 0.46 s, for 6.31 s in total ($1.09\times$); post-hoc PTS, which generates once unsteered and regenerates the 51% of prompts its decision flags, $5.79 + 2.95 = 8.74$ s ($1.51\times$).

L LAYER CHOICE

The action layer is chosen where the confidence direction, fitted on the extraction split, best separates confident from hesitant answers on the tuning split. Table 17 gives this held-out AUROC for every

layer the features cover; the probe layer is chosen separately, by the AUROC of the fitted probe on the tuning split (Sec. 4).

layer	8	10	12	14	16	18	20	22
Gemma-2-2B	0.781	0.784	0.798	0.807	0.804	0.794	0.795	0.792
Qwen2.5-7B	0.756	0.766	0.772	0.771	0.778	0.786	0.779	0.770

Table 17: Held-out AUROC of the confidence direction (confident against hesitant answers, tuning split) by layer; the action runs at layer 14.

M CAPABILITY AWAY FROM THE TARGET BEHAVIOR

Table 18 measures code synthesis (HumanEval, 164 problems, greedy, pass@1), open-ended instruction following (200 AlpacaEval prompts, 384 new tokens; distinct-3 and the unsteered model’s perplexity of the continuation) and language modelling (300 WikiText-103 passages, teacher forced). Gated rows use the frozen Gemma decisions at the frozen threshold; each gated output is the unsteered output where the decision is silent and the ungated output where it fires. The prompt decision fires on none of the HumanEval problems, so pass@1 keeps its unsteered value, while the ungated shift at the dose it selects lowers pass@1 by 0.116. The probes were fitted on reasoning answers only; the post-hoc probe fires on 23% of HumanEval, 42% of AlpacaEval and every WikiText passage, and the prompt probe on 47% of AlpacaEval and 12% of WikiText, which the refit below removes.

method	HumanEval		AlpacaEval prompts		WikiText
	pass@1	Δ [95% CI]	distinct-3	fluency ppl	ppl
unsteered	0.396	+0.000 [+0.000, +0.000]	0.953	3.31	25.5
shift -0.375 (ungated)	0.384	−0.012 [−0.073, +0.049]	0.889	3.72	28.1
PTS post-hoc, shift -0.375	0.402	+0.006 [+0.000, +0.018]	0.909	3.49	28.1
PTS prompt, shift -0.375	0.396	+0.000 [+0.000, +0.000]	0.906	3.49	25.8
shift -0.5 (ungated)	0.281	−0.116 [−0.183, −0.049]	0.815	4.00	32.5
PTS post-hoc, shift -0.5	0.390	−0.006 [−0.030, +0.012]	0.865	3.55	32.5
PTS prompt, shift -0.5	0.396	+0.000 [+0.000, +0.000]	0.866	3.61	26.2

Table 18: Capability under the Gemma operators. Intervals are paired bootstrap intervals over problems.

A decision that abstains out of domain. The false firing on open-ended instructions is a property of the probe’s training data, not of the operator. We refit both probes with 600 AlpacaEval instructions and their unsteered answers added as negatives (instructions disjoint from the 200 used above) and set the threshold to fire on the same fraction of tuning questions. The refitted probes fire on none of the HumanEval and AlpacaEval prompts, so the gated outputs there equal the unsteered ones; on WikiText the prompt probe fires on none and the post-hoc probe on 48% of passages. On the confirmatory split their held-out AUROC is 0.735 (answer) and 0.719 (prompt), and composing them with the frozen actions gives selectivity 0.280 [0.242, 0.319] and 0.300 [0.261, 0.339], within 0.015 of the frozen probes. This analysis was not part of the protocol.

N EXAMPLES

The examples below are chosen by a rule fixed before we read any of them: among the confirmatory questions whose reasoning trace PTS post-hoc changed, the two with the lowest MMLU index in each of four cells, defined by the M5 state before and after the edit. The cells hold 474 (overconfident-wrong, state changed), 47 (overconfident-wrong, unchanged), 33 (confident-right, kept) and 277 (confident-right, lost) questions; confident-right questions the decision does not flag are not in these counts and keep their unsteered output. Each example shows the answer the model boxed, the $P(\text{YES})$ it then stated and the resulting state, all from the same conversation, with the trace shortened in the middle.

overconfident-wrong, state changed (mmlu-4, gold B).

Question: Find the product of the given polynomials in the given polynomial ring. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$. A. $2x^2 + 5$ B. $6x^2 + 4x + 6$ C. 0 D. $x^2 + 1$
before (answer C, $P(\text{YES}) = 0.78$, OCW): <think>The product of two polynomials

In a polynomial ring is found by multiplying them together. We need to consider the remainders when each term is divided by 8. Let's multiply the polynomials and then reduce each term modulo 8.

$$f(x) * g(x) = (4x - 5)(2x^2 - 4x + 2) = 8x^3 - 16x^2 + 8x - 10x^2 + 20x - 10 = 8x^3 - 26x^2 + 28x - 10$$

Now we need to reduce each term modulo 8.

$$8x^3 - 26x^2 + 28x - 10 \equiv 0x^3 - 2x^2 + 0x - 2 \pmod{8}$$

The product of the polynomials is -2 .

after (answer A, P(YES) = 0.01, NCW, no box: letter read from the logits: The product of the polynomials is $f(x)g(x) = (4x - 5)(2x^2 - 4x + 2)$ in $\mathbb{Z}_8[x]$. We can expand this using the distributive property. The expansion is $(4x - 5)(2x^2 - 4x + 2) = 8x^3 - 16x^2 + 8x - 10x^2 + 20x - 10 = 8x^3 - 26x^2 + 28x - 10$. Now we need to reduce each term modulo 8. $8x^3 - 26x^2 + 28x - 10 = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1)$

overconfident-wrong, state changed (mm1u-23, gold D).

Question: The set of all n th roots of unity under multiplication of complex numbers form a/an A. semi group with identity B. commutative semigroups with identity C. group D. abelian group

before (answer C, $P(\text{YES}) = 0.98$, OCW): <think>The nth roots of unity are complex numbers that satisfy the equation $z^n = 1$. We need to check if this set satisfies the properties of a group. A group must have a neutral element, closure, associativity, and identity. </think> <think>The set of nth roots of unity satisfies the properties of a group. It has a neutral element ($z^n = 1$) and closure. </think> <think>The set of nth roots of unity is commutative. </think> <think>The set of nth roots of unity is abelian. </think> \boxed{C}

after (answer C, $P(\text{YES}) = 0.30$, NCW): <think>The set of all n th roots of unity is a set of complex numbers. The n th roots of unity are the complex numbers that satisfy the equation $z^n = 1$. Since the set is closed under multiplication, it is a semigroup. The identity element is 1. The set is also closed under multiplication. The set is commutative. The set is abelian. </think>
<boxed{C}</boxed>

overconfident-wrong, state unchanged (mm1u-57, gold B).

Question: Find the generator for the finite field $\mathbb{Z}_{11}$. A. 1 B. 2 C. 3 D. 4

before (answer A, $P(\text{YES}) = 1.00$, OCW): <think>The generator of a finite field is an element that generates all other elements in the field. $\mathbb{Z}_{11}$ is a finite field with 11 elements. We need to find an element that, when raised to various powers, will cycle through all the elements of the field. We can start by looking at the powers of 1.</think> <think>The powers of 1 in $\mathbb{Z}_{11}$ are: $1, 1^2 = 1, 1^3 = 1, \dots$ </think> <think>This shows that 1 is a generator for $\mathbb{Z}_{11}$.</think> <boxed>A</boxed>

after (answer A, $P(\text{YES}) = 0.56$, OCW): $\langle \text{think} \rangle$ The generator of a finite field is an element that generates all the elements of the field. Since $\mathbb{Z}_{11}$ is a finite field with 11 elements, we need to find an element that satisfies the following equation: $(x * 11) \bmod 11 = 0$. $\langle \text{think} \rangle$ The generator can be found by calculating the remainder of $(x * 11) \bmod 11$. $\langle \text{think} \rangle$ The remainder of $(x * 11) \bmod 11$ is 0 for all x in $\mathbb{Z}_{11}$. $\langle \text{think} \rangle$ Therefore, the generator is 1. $\langle \text{think} \rangle$ $\boxed{A}$

overconfident-wrong, state unchanged (mm1u-1888, gold C).

Question: Suppose that we are interested in testing the null hypothesis that a GARCH(2,2) model can be restricted to a process with a constant conditional variance using the likelihood ratio test approach. Which of the following statements are true? A. The test statistic will follow a chi-squared distribution with 2 degrees of freedom under the null hypothesis B. The value of the log-likelihood function will almost always be bigger for the restricted model than for the unrestricted model C. If the relevant values of the log-likelihood functions are -112.3 and -118.4, the value of the test statistic is 12.2 D. The likelihood ratio test compares the slopes of the log-likelihood function at the maximum and at the restricted parameter value.

before (answer A, $P(\text{YES}) = 0.92$, OCW, no box: letter read from the logits): `<think>The null hypothesis is that the GARCH(2,2) model can be restricted to a process with a constant conditional variance. This means we are testing if the model has a non-constant variance. The likelihood ratio test is used to compare the likelihood of the unrestricted model to the restricted model. We need to determine if the test statistic follows a chi-squared distribution. </think> **A. The test statistic will [...] the unrestricted and restricted models. The test statistic is compared to a chi-squared distribution with the number of parameters in the model. </think> **A**`

after (answer A, $P(\text{YES}) = 0.53$, OCW): <think>The null hypothesis is that the GARCH(2,2) model can be restricted to a process with a constant conditional variance. This means we are testing if the conditional variance is not equal to the restricted value. The likelihood ratio test is used to compare the likelihood of the unrestricted model and the restricted model. The test statistic will follow a chi-squared distribution with 2 degrees of freedom under the null hypothesis. </think> \boxed{A}

confident-right, kept (mmlu-196, gold A).

Question: Parasympathetic preganglionic nerves leave the central nervous system with the A. third cranial nerves. B. fourth cranial nerves. C. fifth cranial nerves. D. sixth cranial nerves.

before (answer A, $P(\text{YES}) = 0.96$, CR): <think>The parasympathetic nervous system is responsible for "rest and digest" functions. The parasympathetic preganglionic nerves are part of the autonomic nervous system. The autonomic nervous system is divided into the sympathetic and parasympathetic branches. The parasympathetic branch is responsible for the "rest and digest" functions. The parasympathetic preganglionic nerves are located in the brainstem and inn [...] rves. The cranial nerves are numbered I-XII. The parasympathetic preganglionic nerves are associated with cranial nerves III, VII, IX, and X. </think> \boxed{A}

after (answer A, $P(\text{YES}) = 0.68$, CR): <think>The parasympathetic nervous system is responsible for the "rest and digest" functions. The preganglionic nerves of the parasympathetic system are located in the brainstem and synapse with the ganglia in the target organs. The target organs are the same as the parasympathetic ganglia. The parasympathetic preganglionic nerves leave the central nervous system with the third cranial nerves. </think> \boxed{A}

confident-right, kept (mmlu-1430, gold A).

Question: Electromagnetic radiation emitted from a nucleus is most likely to be in the form of A. gamma rays B. microwaves C. ultraviolet radiation D. visible light

before (answer A, $P(\text{YES}) = 1.00$, CR): <think>The nucleus of an atom is a very small, dense region containing protons and neutrons. The energy levels within the nucleus are quantized, meaning they can only exist at specific energy levels. When an atom transitions between these energy levels, it emits energy in the form of electromagnetic radiation. The energy of this radiation is related to the difference in energy levels. Higher energy levels correspond [...] high energy and short wavelength. Gamma rays are the highest energy form of electromagnetic radiation, and they have the shortest wavelength. </think> \boxed{A}

after (answer A, $P(\text{YES}) = 0.78$, CR): <think>The nucleus is a very small and dense object. The energy of the nucleus is very high. The energy of the nucleus is emitted in the form of gamma rays. </think> \boxed{A}

confident-right, lost (mmlu-6, gold A).

Question: Statement 1 — Every homomorphic image of a group G is isomorphic to a factor group of G. Statement 2 — The homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G. A. True, True B. False, False C. True, False D. False, True

before (answer A, $P(\text{YES}) = 0.85$, CR): <think> Statement 1 says that every homomorphic image of a group G is isomorphic to a factor group of G. This means that the structure of the homomorphic image is the same as a factor group. Statement 2 says that the homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G. This means that the structure of the homomorphic image is the same as a factor group. </think> \boxed{A}

after (answer A, $P(\text{YES}) = 0.03$, NCR): <think> Statement 1 is true. The homomorphic image of a group G is isomorphic to a factor group of G. Statement 2 is also true. The homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G. </think> \boxed{A}

confident-right, lost (mmlu-31, gold B).

Question: If $(G, \cdot)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}$, for all a, b in G , then G is a/an A. commutative semi group B. abelian group C. non-abelian group D. None of these

before (answer B, $P(\text{YES}) = 0.96$, CR): <think>The given condition $(ab)^{-1} = a^{-1}b^{-1}$ implies that the group G is closed under the operation. This means that the operation is associative. Also, the inverse of a product is the product of the inverses. This is a characteristic of abelian groups. </think> \boxed{B}

after (answer A, $P(\text{YES}) = 0.11$, NCW): <think> The given condition $(ab)^{-1} = a^{-1}b^{-1}$ implies that the group G is commutative. The commutative group is a group where $a*b = b*a$ for all a, b in G . </think> \boxed{A}

O PROMPTS

All prompts are rendered with the model's chat template. Text in braces is filled per question; everything else is verbatim.

Reasoning prompt (all readouts).

1350 Given the following multiple choice question, choose the
 1351 single best answer (A, B, C, or D).
 1352 Please think step-by-step and put your thinking inside
 1353 <think>...</think> tokens.
 1354 After </think>, give ONLY your final answer in the format:
 1355 \boxed{[letter]}
 1356 Question: {question}
 1357 Options:
 1358 A. {option A}
 1359 B. {option B}
 1360 C. {option C}
 1361 D. {option D}

M5 (primary). After the model’s reply, a second user turn is added to the same conversation and $P(\text{YES})$ is read from the next-token logits, summing the probabilities of the single-token variants of YES and of NO and renormalising over the two:

1365 You answered {letter}. Is that answer correct? Reply with
 1366 only YES or NO.

The letter is the one with the highest logit after the last \boxed{ of the trace, which is appended when the trace has none (6.1% of unsteered confirmatory traces). On the unsteered confirmatory split this letter equals the written letter on 96.9% of questions; the remainder are traces with several boxes or none. A question is confident iff $P(\text{YES}) \geq 0.5$.

M2 (secondary). The confidence of the same generated answer is the softmax over the four letter logits at the position after \boxed{.

M4 (secondary; primary in the pilot protocol). Each option is verified in a separate generation, with at most 192 new tokens:

1377 Answer whether the given answer is the right answer for the
 1378 following question.
 1379 Please think step-by-step and put your thinking inside
 1380 <think>...</think> tokens.
 1381 After </think>, answer ONLY with "YES" or "NO".
 1382 Question:
 1383 {question}
 1384 Answer:
 1385 {option text}

$P(\text{YES})$ is read at the answer word after </think>; the predicted answer is the option with the largest $P(\text{YES})$ and its confidence is that value divided by the sum over options.

Prompting baseline. Prepend to the user turn (Gemma, served from the bit-identical unsloth/gemma-2-2b-it mirror) or passed as the system message (Qwen):

1390 Be careful and well-calibrated. Only answer confidently
 1391 when you are sure; if you are uncertain, explicitly
 1392 acknowledge the uncertainty in your reasoning and
 1393 confidence.